# Attention and Regret*

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**Abstract**

This paper develops an optimality foundation for regret theory. We ask which emotional responses to decision outcomes would best incentivize an agent to pay more attention. Regret emerges as the uniquely optimal emotion. Our approach provides a criterion to arbitrate between different versions of regret theory and generates new predictions. Regret depends on decision complexity and exhibits a self-blame component: regret following mistakes is stronger in simpler decision problems.

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*[Regret] may also depend on the extent to which the individual blames himself for his original decision. [...] The neglect of this dimension of regret—although a useful simplifying assumption for many problems—is a serious obstacle to the development and generalisation of regret theory.* - Sugden 1986

# I Introduction

Decisions under uncertainty can trigger regret or rejoicing when one learns that another action would have yielded better or worse outcomes. These emotions are often viewed as futile "crying over spilled milk," and their anticipation has been shown to distort choices, violating expected utility theory (Bell 1982; Loomes and Sugden 1982; Zeelenberg 1999a). This raises a puzzle: evolutionary fitness depends on actual outcomes, not counterfactual ones, and these emotions distort choices. Why, then, do regret and rejoicing exist?

Although the anticipation of regret and rejoicing can distort decision-making, these emotions can nevertheless improve the quality of decisions if we take into account attention. When facing uncertainty, individuals can improve decision-making by paying attention, which we use as an umbrella term for both external information acquisition and internal information processing such as deliberation, planning, or memory search. The anticipation of regret following mistakes should motivate individuals to deliberate more carefully. This mechanism is empirically supported by Reb (2008), who finds that making regret salient leads subjects to take longer to reach a decision and to acquire more information. Regret can thus be seen as a penalty for not thinking hard enough, in line with the idea that regret has a self-blame component.

We hypothesize that regret is culturally instilled to motivate more attention and thereby improve decision-making. Just as guilt is instilled to discourage immediate gratification and antisocial behavior (Tracy, Robins, & Tangney 2007), we assume regret is instilled to encourage deliberation in decisions under uncertainty. The mechanism we consider is parental socialization: when parents consistently blame or praise children for certain outcomes, parental evaluations become internalized as self-blame or self-congratulation over time (Hoffman 1983; Kochanska 1993). This view aligns with the economics literature on intergenerational preference transmission, which emphasizes that parents actively shape children's preferences and values (Bisin & Verdier 2011; Doepke, Sorrenti, & Zilibotti 2019). Parents want children to pay more attention than children would themselves when there is disagreement in time preferences: attention costs are borne in the present while decision outcomes materialize in the future, so parents want more deliberation if they are more patient than their children or if children exhibit present bias.

To investigate our hypothesis that regret evolved to motivate attention, we ask whether regret is the optimal emotion for this purpose. This is not clear ex ante because any emotion that rewards good outcomes and penalizes bad outcomes would arguably raise the stakes of decisions and thereby motivate attention. For instance, this would also be achieved by disappointment and elation, which also depend on counterfactual comparisons, but to other states of the world rather than other actions. Using a mechanism-design approach, we show that regret and rejoicing emerge as optimal from a non-parametric class of emotions and that their distinctive structure—counterfactual comparisons to alternative actions—is what makes them optimal for motivating attention. We then leverage our functional approach to deliver a model of regret. This model connects closely to existing regret theories while enriching them by explicitly accounting for attention.

We formalize this hypothesis in the language of the principal-agent approach to the evolution of preferences.[1] A parent (principal) shapes a child's (agent's) emotional responses to incentivize attention. The child is biased toward insufficient attention due to disagreement in time preferences, as described above. The parent chooses the child's emotional utility as a function of the chosen action and realized state. Emotional utility cannot condition (i) on attention, inducing a hidden-action problem, or (ii) on the child's bias, inducing a hidden-information problem. The first constraint captures that attention is internal to the child's mind, while decision outcomes are observable. The second reflects that emotions are socialized in early childhood, when preferences are malleable but not yet stable; parents cannot target the exact bias their child will eventually have. We model attention using the rational-inattention paradigm, where the agent can choose any signal structure subject to an entropy-based information cost (Sims 2003). This creates a high-dimensional mechanism-design problem with both hidden information and hidden actions. To render the model tractable, we restrict ourselves to decision problems that are invariant under relabeling actions and impose that emotional utility preserves this symmetry.

Regret and rejoicing are optimal because they amplify utility differences between actions depending on the material stakes in each state. Formally, let $v(a, \omega)$ denote the material payoff under action $a$ and state $\omega$. In our model, regret or rejoicing experienced in state $\omega$ under action $a$ is a function of the vector of payoff differences $v(a, \omega) - v(a', \omega)$ to alternative actions $a'$ under the realized state $\omega$. By rewarding better actions and penalizing worse actions, regret and rejoicing amplify the state-wise utility differences between actions, which we refer to as the *stakes* in a given state. The emotional amplification of stakes is optimal because it motivates the agent to pay more attention to the state. Regret and rejoicing amplify these stakes by an amount that depends on the material payoff differences between actions, that is, the material stakes. By depending on material stakes, these emotions optimally direct attention to states where stakes are higher.

For binary decision problems, the canonical setting in regret theory, we obtain sharper results. The optimal emotions yield *original regret theory* (Bell 1982; Loomes and Sugden 1982), including its key assumption, disproportionate aversion to large regrets. This assumption means that regret amplifies larger utility differences between actions disproportionally. Disproportionate aversion to large regrets explains well-documented violations of expected utility theory, including the Allais paradox, simultaneous gambling and insurance, and preference reversals (Bell 1982; Loomes and Sugden 1982). Our model predicts a weaker version of this property—starshapedness rather than convexity—but we demonstrate that starshapedness is sufficient to generate the above violations

[1]The principal-agent approach to the evolution of preferences has been employed, among others, by Robson (2001b), Samuelson and Swinkels (2006), Rayo and Becker (2007), Netzer (2009), and Baliga and Ely (2011).

of expected utility that motivated the convexity assumption.

In our model, disproportionate aversion to large regrets arises from heterogeneity in the agent's bias toward insufficient attention. When the material stakes are already large, agents with small biases choose the ex-post optimal action most of the time, and their mistake rates are relatively insensitive to further increases in the stakes. Agents with stronger biases make more frequent mistakes, and their behavior is more responsive to stake increases. The principal therefore tailors the emotional amplification to the more biased types by scaling up larger material stakes disproportionately more.

Original regret theory has two limitations that our model addresses: it does not account for decision complexity and was developed only for binary choice.

First, our model predicts that regret depends on the complexity of the decision problem. As emphasized by Sugden (1986) and decision-justification theory (Connolly & Zeelenberg 2002), regret has a self-blame component, and self-blame depends on how justifiable a decision was. A mistake is less justifiable if the decision problem was easy, so simpler decision problems should elicit more regret following mistakes. Our model features a complexity parameter $\lambda$ that captures how costly it is to discover the right action, and indeed predicts that simpler problems elicit stronger regret. Our model also predicts that regret-based distortions to choice and attention vanish as decision problems become very simple or very complex.

Extending regret theory to incorporate decision complexity raises concerns about introducing new parameters. One might expect a different regret function for each complexity level, requiring separate measurement for each, which would limit the model's applicability and falsifiability. A virtue of our microfoundation is that it disciplines this extension. A scaling property implies that regret at any complexity can be derived from regret at a reference complexity via a simple transformation, so no additional preference parameters are needed. The complexity parameter can be identified from the stochasticity of choice, following standard methods in rational inattention (Caplin 2016).

Second, our model predicts regret beyond binary choice, which has proven difficult for original regret theory (Loomes and Sugden 1982; Loomes and Sugden 1987). A frequently used model for general choice sets considers only regret relative to the ex-post best action (Quiggin 1994). Our model makes the psychologically plausible prediction that regret and rejoicing are affected by all counterfactual actions, in line with generalized regret theory (Loomes and Sugden 1987; Sugden 1993). Our model strengthens generalized regret theory by predicting that regret depends only on material utility differences and that regret maintains the utility ordering of actions.

Our results have testable implications for feedback interventions. Regret is mediated by counterfactual feedback: whether decision-makers learn ex post what would have happened under alternative choices (Zeelenberg 1999a). Previous work has shown that if subjects anticipate counterfactual feedback, this induces choice distortions in line with regret theory (Filiz-Ozbay & Ozbay 2007; Zeelenberg 1999a). Our paper emphasizes that counterfactual feedback also affects attention: providing feedback increases attention by inducing regret but distorts attention toward high-stakes states. Moreover, these effects depend on decision complexity. The incentive effects of regret are strongest for simpler problems (Proposition 3), while attention and choice distortions peak at medium complexity (Proposition 4). These results can inform the optimal use of feedback interventions, which are cheap ways to incentivize attention but generate predictable distortions.

**Related Literature** We contribute to the large literature on regret in economics, starting with Bell (1982) and Loomes and Sugden (1982). We introduce their model, original regret theory, in section A.1. Bleichrodt and Wakker (2015) review empirical support and applications of regret theory across domains such as finance, health, insurance, and auctions. We provide an optimality foundation for original regret theory that simultaneously extends it to incorporate decision complexity.

Our approach follows the shift in emotions research toward functional accounts of emotions (Keltner & Gross 1999), and shares with cultural constructionist accounts an emphasis on cultural influences on emotions (Mesquita & Parkinson 2024). Psychologists often think of regret as an adaptive emotion that improves decision-making (e.g., Janis and Mann 1977; Zeelenberg 1999b; O'Connor, McCormack, and Feeney 2012). This view is supported by Camille et al. (2004), who find that patients with damage to their orbitofrontal cortex, who do not experience regret but are otherwise cognitively intact, make worse decisions. We provide an attention-based mechanism through which regret improves decisions. Independently, Compte (2025) develops a model where an internal regulator is influenced through psychosomatic costs, with an application to regret as a way to motivate attention. His model restricts to negative emotions, assumes emotional costs are borne by both selves, and allows for asymmetric decision problems. The attention mechanism is empirically supported by Reb (2008), who finds that making regret salient leads subjects to take longer to reach decisions and to acquire more information.

A popular alternative hypothesis for how regret may benefit decision-making is that it improves learning in repeated decisions. When agents experience regret, they purportedly learn to modify their future behavior. This hypothesis draws theoretical support from reinforcement learning with regret, called regret matching, which has desirable asymptotic properties (Foster and Vohra 1999).[2] However, regret in reinforcement learning and regret in decision theory are very different. Regret matching predicts stochastic behavior where the probability of an action is proportional to its historical regret. By contrast, regret theory predicts agents choose the action with the highest expected utility, where utility has a regret component. Our framework provides an optimality foundation that delivers the behavioral predictions of regret theory (see Section A.3).

This paper relates to the economics literature on intergenerational preference transmission (Bisin & Verdier 2011; Doepke, Sorrenti, & Zilibotti 2019). Our approach is closest to models where parents actively shape children's preferences, as in Bisin and Verdier (2001) and Doepke and Zilibotti (2017), rather than models of cultural transmission according to certain rules, such as imitation of successful individuals (Boyd & Richerson 1985). We contribute to this literature by

[2] Regret also appears prominently in reinforcement learning as a performance benchmark for learning algorithms ("no regret learning"). Regret matching is one of many learning algorithms that achieves this benchmark.

using the principal-agent approach to the evolution of preferences (e.g., Robson 2001a; Rayo and Becker 2007) to allow parents to choose children's utility function in a more flexible way. Instead of choosing a preference parameter, parents choose the entire utility function over decision outcomes, justified by the technology of emotion internalization: parents can blame or praise children for any outcome, and these evaluations become internalized as part of the utility function.

At a broader level, our paper contributes to the literature on how a principal should incentivize an expert to acquire information, beginning with Demski and Sappington (1987). Our model allows for flexible information acquisition, as in recent contributions such as Lindbeck and Weibull (2020) and Yoder (2022), and is distinctive in combining this with both hidden actions (signal structures are not contractible) and hidden information (the agent's bias is private). We contribute to this literature by showing that counterfactual comparisons (as they factor into regret) are useful for incentivizing information acquisition.

Yoder (2022) also considers flexible information acquisition and hidden information, but assumes that either the signal structure or the signal realization is contractible. In contrast, in our model only the action and realized state are contractible. We show that self-screening is ineffective and hence our solution methods and the resulting distortions differ: in his model, there is no distortion at the top, whereas in our case, information acquisition is skewed towards extreme states for all types. Methodologically, Yoder (2022) uses the posterior approach (Caplin, Dean, & Leahy 2022), while we use the state-dependent choice probabilities approach (Matějka & McKay 2015) and its connection to perturbed utility models (Bloedel, Denti, & Pomatto 2025).

The paper is organized as follows. Section II introduces and discusses the model. Section III presents the results under two and more than two actions. Section IV concludes. The Appendix provides proofs and extensions. Appendix A contains proofs. Appendix B shows how to extend the model to incomplete ex-post feedback about the state of the world, where disappointment and elation arise to supplement regret and rejoicing to motivate attention. Appendix C discusses how our model relates to more standard principal-expert problems in economics.

# II Model

A principal (she) shapes the preferences of an agent (he) who faces a decision problem under uncertainty. The agent can pay attention to the state of the world before choosing an action, but is biased toward insufficient attention from the principal's perspective. The principal instills emotions as a function of the chosen action and the realized state. Our primary interpretation is cultural evolutionary: parents (principal) shape children's (agent) emotional responses through evaluations that become internalized over time.

**Decision Problem** A *decision problem* is a tuple $(\Omega, \mu, A, v)$. It comprises a finite state space $\Omega$, a full-support prior $\mu \in \Delta(\Omega)$ on the state space, a finite set of actions $A$, and a material utility function $v\colon A \times \Omega \to \mathbb{R}$.[3]

**Attention** The agent faces a two-stage problem. First, he acquires a signal structure about the state (attention stage); second, he chooses an action from $A$ conditional on the realized signal (decision-making stage). A signal structure is a pair $(S, \sigma)$, where $S$ is a finite signal space and $\sigma\colon \Omega \to \Delta(S)$ is a state-dependent distribution over signals, with $\sigma(s|\omega)$ denoting the probability of signal $s$ in state $\omega$. Following the rational-inattention approach (Sims 2003), the agent can select any signal structure subject to the mutual information cost[4]

$$c(S,\sigma) := \sum_{s\in S}\Big(-\sigma(s)\log\sigma(s)\Big) - \sum_{\omega\in\Omega}\left(\mu(\omega)\sum_{s\in S}\Big(-\sigma(s|\omega)\log\sigma(s|\omega)\Big)\right),$$

where $\sigma(s) := \sum_{\omega\in\Omega}\mu(\omega)\sigma(s|\omega)$ is the unconditional probability of signal $s$ and log refers to the natural logarithm (with $0\log 0 := 0$). The first term is the entropy of the unconditional signal distribution; the second is the expected entropy of the signal conditional on the state. Entropy captures the uncertainty of a distribution, so the cost reflects the reduction in uncertainty about the signal from knowing the state—that is, how much the signal is tailored to the state.

As is well known, it suffices to consider signal structures that recommend an action, i.e., $S = A$. The reason is that mutual information is Blackwell monotonic, so the agent benefits from garbling any signal structure into an action recommendation (Matějka & McKay 2015). This reduces the agent's two-stage problem to selecting state-dependent choice probabilities $p\colon \Omega \to \Delta(A)$, written $p(a|\omega)$, which we call an *attention strategy.* Writing $c(p)$ for $c(A, p)$, the attention cost captures how much the action is tailored to the state.

**Principal's Preferences** The principal derives utility $V(p)$ from attention strategy $p$:

$$V(p) = \sum_{\omega\in\Omega}\sum_{a\in A}\mu(\omega)p(a|\omega)v(a,\omega) - \lambda c(p).$$

The principal internalizes the attention cost weighted by $\lambda$ and internalizes the agent's material utility $v(a,\omega)$ from action $a$ under state $\omega$. We interpret $\lambda$ as the complexity of the decision problem, since it scales the cost of learning about the state.

**Agent's Bias** The agent has a bias $\beta \in (0,1)$ that discounts the material payoff $v(a,\omega)$, or, equivalently, overweights the attention cost by $1/\beta$. Under our interpretation, parents want children to deliberate more carefully than children do. Since attention is a costly activity whose payoff realizes only later, we interpret this as disagreement about patience. Such disagreements are

[3]We assume a finite state space to circumvent measure-theoretic issues. We believe our results extend to infinite state spaces under appropriate technical assumptions.

[4]While we assume the canonical mutual information cost, which has foundations in information theory (Shannon 1948), we expect most of our results to hold for the broader class of Csiszár information costs (Bloedel, Denti, & Pomatto 2025). These costs satisfy the state-separability property that drives the main results.

central to the literature on intergenerational preference transmission, which has studied how parents socialize children to value future-oriented investments such as hard work and education (Lindbeck and Nyberg 2006; Doepke and Zilibotti 2017).

The disagreement between parents and children can arise from two sources. First, parents are more patient than children and have paternalistic preferences: they use their own preferences to evaluate children's choices (Bisin & Verdier 2001; Doepke & Zilibotti 2017). Second, the child is tempted to shirk on attention due to present bias, and parents want to correct for this bias. Present bias is one of the most studied biases in behavioral economics, with substantial evidence for its prevalence (Imai, Rutter, & Camerer 2021). Under this interpretation, regret serves as an internal self-control mechanism for paying sufficient attention.

The bias $\beta$ is heterogeneous and private information of the agent, with cumulative distribution function $F$. We make two assumptions on the distribution of the bias $\beta$. First, the bias is bounded away from zero, which ensures that finite emotions are optimal. Second, $\log \beta$ admits a strictly log-concave density, which is used to show uniqueness and holds if $\beta$ admits a log-concave density.

**Assumption 1.** *The bias $\beta$ has support $\operatorname{supp}(\beta) = [\underline{\beta}, 1]$ where $\underline{\beta} > 0$. The distribution of $\log \beta$ admits a strictly log-concave density.*

Heterogeneity of preferences and biases across individuals is well documented. Under the interpretation that parents are more patient than children and impose their own time preferences (paternalism), heterogeneity in $\beta$ reflects heterogeneity in children's patience—more impatient children have a larger bias from the parent's perspective. Studies consistently find substantial variation in time preferences within populations (Frederick, Loewenstein, & O'Donoghue 2002). Under the interpretation that parents correct for children's present bias, heterogeneity in $\beta$ arises because present bias varies across individuals (Augenblick & Rabin 2019).

The bias is private information of the agent. While parents may have better information about their children's preferences than the population distribution, emotions are socialized in early childhood (Eisenberg, Cumberland, & Spinrad 1998; O'Connor, McCormack, & Feeney 2012), when preferences are malleable but not yet stable. Because emotional responses must be instilled before preferences have fully formed, parents cannot target the specific $\beta$ their child will eventually have. The distribution $F$ captures this residual uncertainty about the child's realized bias. Under the present-bias interpretation, an additional source of uncertainty arises: self-control varies from situation to situation. Cognitive load reduces self-control (Shiv & Fedorikhin 1999), as do stress and cues that trigger visceral urges (Loewenstein 1996). Thus, even if a parent could perfectly predict their child's baseline patience, the effective bias in any given decision remains stochastic.[5]

[5] There is also a strategic rationale for why tailoring emotions to $\beta$ is difficult even if $\beta$ is permanent. Lemma 1 shows that if parents conditioned emotions on the child's reported bias, this would not benefit parents: higher types prefer stronger incentives (due to their higher return to attention) and select into them, while lower types select weaker incentives. This is the opposite of what the principal wants, so screening backfires. The same adverse-selection logic applies if parents use the children's behavior to infer $\beta$ and tailor emotions. Children would have incentives to manipulate their behavior to obtain their preferred incentive scheme, limiting the scope for tailoring emotions.

**Emotions** The principal shapes the agent's preferences over decision outcomes by choosing an emotion-inclusive utility function $u : A \times \Omega \to \mathbb{R}$. We interpret $m(a, \omega) := u(a, \omega) - v(a, \omega)$ as emotional utility, which can depend on the chosen action and the realized state.[6] The agent with bias $\beta$ derives utility $U_\beta$ from the attention strategy $p$,

$$U_\beta(p) = \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) u(a, \omega) - \lambda c(p).$$

The underlying technology is that parental evaluations of children's observed behavior become internalized over time. When parents consistently blame or praise children in response to certain outcomes, children learn to anticipate these reactions, which eventually become self-generated emotional responses. This mechanism is central to theories of moral internalization (Hoffman 1983) and conscience development (Kochanska 1993), where parental evaluations become internal self-evaluations (Tangney & Dearing 2002).

Crucially, emotions can only condition on observable behavior—the chosen action and the realized state—not on the attention paid. This reflects that attention is largely internal to the child's mind, while decision outcomes are observable. This constraint is why regret emerges as optimal: the principal cannot directly reward attention, only outcomes.[7]

We also assume the principal cannot directly modify the child's weight on attention costs. While parents may teach patience to some degree, evidence on the persistence of present bias suggests limits to this strategy. Moreover, even if parents could uniformly scale up the child's weight on outcomes, this would be dominated by shaping state-dependent preferences over outcomes, which allow finer tailoring of incentives (see Theorem 1).[8]

**Socialization Cost as Tie-Breaker** The model so far identifies optimal emotions only up to state-wise constants. The reason is that the principal cares only about the agent's behavior, and shifting all emotional utilities in a given state by a constant does not alter behavior because the agent is an expected-utility maximizer.[9] For readers interested only in behavioral predictions, this indeterminacy is immaterial. For the interpretation of $m = u - v$ as experienced emotions, state-wise constants determine whether emotions are positive or negative. To resolve this indeterminacy, we introduce a socialization cost (Bisin and Verdier 2001; Doepke and Zilibotti 2017).

In case of indifference, the principal selects the utility function $u$ that minimizes the expected

[6] Additive separability of emotions in utility is standard (Bell 1982, 1985; Loomes and Sugden 1982, 1986).

[7] If the principal could condition on attention, she could achieve the first best by penalizing deviations from the optimal attention strategy. Similarly, if the bias were observable and contractible, she could achieve the first best by scaling up payoffs by $1/\beta$.

[8] Formally, increasing $\beta$ uniformly by factor $k$ replicates scaling stakes by $k$. But optimal emotions scale stakes non-linearly and state-dependently.

[9] As in regret theory, *given emotions*, our agent is an expected-utility maximizer. However, because emotions can be a function of the choice set, the agent's choice behavior can violate expected-utility axioms. To be precise, that state-wise constants do not affect behavior also relies on the information cost being additively separable, so there are no "income effects" on attention from better emotions.

socialization cost

$$\Phi(m) := \int \left( \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) \phi\big(u(a,\omega) - v(a,\omega)\big) \right) dF(\beta),$$

where $m(a,\omega) = u(a,\omega) - v(a,\omega)$ is the internalized emotion and $\phi(m)$ is the socialization cost of making children internalize this emotion. Both positive and negative emotions are costly to socialize. We assume $\phi$ is strictly convex, differentiable, and minimized at 0.

**Assumption 2.** *The socialization cost $\phi\colon \mathbb{R} \to \mathbb{R}$ is strictly convex, differentiable, and minimized at 0.*

This tie-breaker can be understood as the limit of a model where the expected socialization cost $\Phi$ enters the principal's objective additively with vanishing weight relative to $V(p)$. If socialization costs entered the objective, the optimal emotion would still have the structure of regret, but the shape of regret—its magnitude and sensitivity to stakes—would be affected by the socialization cost $\phi$. We adopt the limiting case to ensure that emotions are shaped by the optimal incentives for attention rather than the functional form of $\phi$.

The principal's objective is material payoff $v$, not emotion-inclusive utility $u$. Following the emotion-socialization literature (Hoffman 1983; Kochanska 1993), we treat emotions as tools to shape behavior rather than as sources of well-being to be maximized. This is consistent with the prominent role of negative emotions—guilt, shame, regret—in emotional development, which suggests that parents do not strongly internalize children's emotional experiences. This assumption can be relaxed. Suppose parents partially internalize children's emotional well-being, so the effective cost becomes $\tilde{\phi}(m) = \phi(m) - \alpha m$ for some $\alpha > 0$. If $\phi(m) = |m|$ and $\alpha < 1$, then $\tilde{\phi}$ is still minimized at zero, leaving the analysis unchanged. For differentiable $\phi$, the cost $\tilde{\phi}$ is minimized at some $a > 0$ rather than at zero. This is equivalent to a model with some cost function $\hat{\phi}$ satisfying our assumptions, combined with a uniform upward shift in all emotions by $a$. The structure of optimal emotions is preserved; only the level changes.

**Symmetry** The presence of both hidden actions and hidden information creates a high-dimensional mechanism-design problem. In particular, the agent's attention strategy generally does not have a closed-form solution. To render the problem tractable, we make two symmetry assumptions. First, we impose that the material utility $v$ is symmetric in the actions as defined below. Second, we restrict ourselves to emotions that maintain this symmetry. Under symmetric utility, the agent's attention strategy takes the multinomial logit form (Matějka & McKay 2015); without symmetry, the solution involves action biases that depend on the entire decision problem and generally lack closed-form expressions.

**Definition 1.** *A function $w\colon A \times \Omega \to \mathbb{R}$ is **symmetric** if the distribution over payoff vectors $(w(a,\omega))_{a \in A} \in \mathbb{R}^A$ induced by prior $\mu$ is invariant under permutation of the coordinates.*

Symmetric utility means that the decision problem is invariant under relabeling the actions. As an example, the utility function according to the payoff-matrix in Table 1 is symmetric if $\mu(\omega_1) = \mu(\omega_2)$ and $\mu(\omega_3) = \mu(\omega_4)$. That is because the utility vector $(0, 1)$ is as likely as its permutation $(1, 0)$ and the utility vector $(3, 2)$ is as likely as $(2, 3)$.

| | $\omega_1$ | $\omega_2$ | $\omega_3$ | $\omega_4$ |
|---|---|---|---|---|
| $a_1$ | 1 | 0 | 3 | 2 |
| $a_2$ | 0 | 1 | 2 | 3 |

Table 1: Symmetric payoff matrix

**Assumption 3.** *The material utility $v$ is symmetric.*

This assumption is plausible when the actions are indistinguishable before information processing (Matějka & McKay 2015), a natural benchmark for situations where all actions are "plausible candidates for choice" (Sugden 1986) and the agent has no strong prior intuition. This describes typical applications of regret theory in finance, insurance, and health, where the decision-maker chooses among assets, insurance contracts, or medical procedures without an ex-ante favorite.

For tractability, we focus on emotion-inclusive utility functions $u$ that maintain symmetry. Thus, the emotions do not depend on the labeling of actions, but we otherwise allow for a large non-parametric class of emotions, including disappointment and elation (Loomes & Sugden 1986). This symmetry assumption is natural because the underlying decision problem is symmetric in the actions, so the first-best attention strategy is symmetric in the actions.[10]

**Principal's Problem** Combining the above, the principal solves the following problem (P).

$$\max_{u,(p_\beta)_\beta} \int \left( \sum_{\omega\in\Omega} \sum_{a\in A} \mu(\omega) p_\beta(a|\omega) v(a,\omega) - c(p_\beta) \right) dF(\beta) \tag{P}$$

s.t.

$$\forall\beta\colon\ p_\beta \in \arg\max_p \beta \sum_{\omega\in\Omega} \sum_{a\in A} \mu(\omega) p(a|\omega) u(a,\omega) - c(p) \tag{IC}$$

$$u \text{ is symmetric.} \tag{SYM}$$

Out of the solutions to (P), the principal chooses the one that minimizes the socialization cost $\Phi(u)$.

One may wonder whether the principal could tailor emotions to the agent's type by asking the agent to report their bias. However, such screening is typically valuable only when the principal wants different types to take different actions. Here, the principal wants the agent to pay the same amount of attention regardless of the agent's bias. In Appendix A, we formally show that under

[10]Simulations under two states suggest that inducing asymmetries through emotions cannot benefit the principal. For a quadratic posterior-separable information cost under two states, one can show that asymmetric utility is suboptimal.

two states, screening cannot improve upon inculcating a single emotion, so the formulation of (P) is without loss. In fact, any attempt to screen backfires: higher types select stronger incentives while lower types select weaker incentives, increasing the variance of attention.

# III Results

First, we characterize the solution of (P) for decision problems containing two actions and then turn to the case of $n > 2$ actions. Assumptions 1 to 3 are maintained throughout.

## A Two Actions

This section considers binary decision problems, $A = \{a_1, a_2\}$. We first introduce original regret theory, then characterize the optimal emotion in our model, and finally compare our solution to existing regret theories.

### A.1 Original Regret Theory

To verify that the optimal emotion of our model can be interpreted as regret and rejoicing, we introduce *original regret theory* due to Bell (1982) and Loomes and Sugden (1982).

The idea behind original regret theory is that upon learning the realized state, agents experience not only realized payoff but also emotional utility depending on the payoff difference to the counterfactual action. When choosing between two actions, agents compare expected emotion-inclusive utility. Formally, let $\mathcal{A}$ be a grand set of actions. Original regret theory assumes a material utility function $v : \mathcal{A} \times \Omega \to \mathbb{R}$, a regret-rejoicing function $R : \mathbb{R} \to \mathbb{R}$, and a subjective belief $\pi \in \Delta(\Omega)$. Action $a_i$ is weakly preferred to action $a_j$, written $a_i \succsim a_j$, if and only if

$$\sum_{\omega\in\Omega} \pi(\omega)\Big(v(a_i,\omega) + R\big(v(a_i,\omega) - v(a_j,\omega)\big)\Big) \geq \sum_{\omega\in\Omega} \pi(\omega)\Big(v(a_j,\omega) + R\big(v(a_j,\omega) - v(a_i,\omega)\big)\Big). \quad (1)$$

Here, $R(x)$ is the emotional utility from a material utility difference $x$ to the counterfactual action: $R(x) < 0$ if $x < 0$ (regret) and $R(x) > 0$ if $x > 0$ (rejoicing).

While intuitive, the representation in terms of $R$ contains redundancy: different functions $R$ can induce identical preferences. The literature therefore typically works with the following equivalent formulation. Subtracting the right-hand side from the left-hand side of (1), we obtain

$$a_i \succsim a_j \Leftrightarrow \sum_{\omega\in\Omega} \pi(\omega) Q\big(v(a_i,\omega) - v(a_j,\omega)\big) \geq 0, \quad (2)$$

where $Q(x) := x + R(x) - R(-x)$ is the emotion-inclusive utility difference between actions in a state where the material utility difference is $x$. By construction, $Q$ is skew-symmetric, $Q(-x) = -Q(x)$.

Equation (2) shows that $Q$ captures all behavioral consequences of regret. Original regret theory is defined as preferences of the form (2) with skew-symmetric $Q$. Diecidue and Somasundaram

(2017) axiomatize this representation, and Bleichrodt, Cillo, and Diecidue (2010) show how to identify $v$ and $Q$ from choices.

Under non-linear $Q$, regret theory allows for violations of expected utility theory such as non-transitive preferences. Bell (1982) and Loomes and Sugden (1982) observe that if $Q$ is strictly convex, regret theory explains the Allais paradox, the common-ratio effect, and simultaneous gambling and insurance.

### A.2 Optimal Emotion

We now define what it means for the solution to the principal's problem to take the form of regret theory. Our definition follows original regret theory but requires only that $Q$ is starshaped rather than convex.

**Definition 2.** *A function $g : \mathbb{R} \to \mathbb{R}$ is **starshaped** if $g(0) = 0$, $g(x) > 0$ for $x > 0$, and if for all $x > 0$ and $k > 1$:*

$$g(kx) > k\, g(x).$$

This property is also known as strictly increasing returns to scale in the context of production functions. Our definition of starshapedness lies between strict convexity and strict superadditivity for nonnegative functions on $[0, \infty) \to \mathbb{R}$ that vanish at 0 (Bruckner & Ostrow 1962).

For $a_i$, $i = 1, 2$, denote by $a_{-i}$ the other action in $A$, that is, $a_{-1} = a_2$ and $a_{-2} = a_1$.

**Definition 3.** *Under two actions, $A = \{a_1, a_2\}$, a solution $(u, (p_\beta)_\beta)$ to the principal's problem* (P) *is a **regret solution** if there exists a continuous function $R\colon \mathbb{R} \to \mathbb{R}$ such that for all $(a_i, \omega) \in A \times \Omega$:*

$$u(a_i, \omega) = v(a_i, \omega) + R\big(v(a_i, \omega) - v(a_{-i}, \omega)\big)$$

*and*

1. *$R(x) \gtreqless 0 \Leftrightarrow x \gtreqless 0$,*
2. *$Q(x) := x + R(x) - R(-x)$ is starshaped.*

Note that a regret solution requires not only that the agent's preferences can be represented as in regret theory, but that the emotional utility $m$ itself takes the form of the regret-rejoicing function in Bell (1982) and Loomes and Sugden (1982).

Property 1 states that emotional utility is positive (rejoicing), zero, or negative (regret) according to whether the counterfactual material utility is worse, equal to, or better than the actual material utility. The requirement that $R$, and thus $Q$, is continuous is important for measurement (Diecidue and Somasundaram 2017). Property 2 requires that regret and rejoicing increase more than proportionally as a function of the material utility difference, which we also refer to as disproportionate aversion to large regrets. Section A.3 shows that starshaped $Q$ delivers the behavioral predictions of regret theory.

The following theorem shows that the unique solution to the principal's problem is a regret solution. This is perhaps surprising: one might expect the optimal emotion to depend on the entire decision problem, that is, all material utilities and the prior. Instead, emotional utility depends only on the material utility difference between the chosen and counterfactual action in the realized state. Moreover, the regret-rejoicing function $R$ is independent of the decision problem, though it depends on $\lambda$ and the distribution of $\beta$, which we use for comparative statics below.

**Theorem 1.** *Under a binary decision problem, the unique solution to the principal's problem* (P) *is a regret solution, where $R$ is independent of the decision problem.*

The theorem states that optimal emotions take the form of original regret theory. The intuition for why a regret solution is optimal is as follows.

Recall that the principal seeks to motivate more attention from the agent. Define the *emotion-inclusive stakes* $\Delta u(\omega) := u(a_1, \omega) - u(a_2, \omega)$ as the emotion-inclusive utility difference between actions. The key observation is that these stakes determine how much attention the agent pays to a state. Under rational inattention, the agent's choice probability is given by a logit function (Matějka & McKay 2015):

$$p_\beta(a_1|\omega) = \frac{e^{\frac{\beta}{\lambda}\Delta u(\omega)}}{1 + e^{\frac{\beta}{\lambda}\Delta u(\omega)}} \tag{3}$$

Higher emotion-inclusive stakes lead to a higher probability of choosing the correct action. Regret and rejoicing motivate attention by increasing these stakes. Property 1 reflects this: regret penalizes the wrong action and rejoicing rewards the right action.

The preceding explains why emotions raise the stakes. But why do they depend on counterfactual comparisons—the difference to what one could have gotten? Define the *material stakes* as $\Delta v(\omega) := v(a_1, \omega) - v(a_2, \omega)$. The material stakes capture the value of getting the decision right in a state and thus determine how much attention the principal wants the agent to pay. Optimal emotions are a function of the material stakes, to align the agent's attention with what is optimal.

The map from material to emotion-inclusive stakes is therefore the key object the principal chooses. This map is $Q$: given material stakes $\Delta v$, the better action yields rejoicing $R(\Delta v)$ while the worse action yields regret $R(-\Delta v)$, so emotion-inclusive stakes are $\Delta v + R(\Delta v) - R(-\Delta v) = Q(\Delta v)$. We now turn to the shape of $Q$.

Starshapedness of $Q$ stems from uncertainty about the bias $\beta$. If $\beta$ were known, the optimal $Q$ would be linear, scaling up the stakes by the inverse of the bias:

$$Q(\Delta v) = \frac{\Delta v}{\beta}. \tag{4}$$

This would perfectly offset the agent's bias, implementing the first-best attention strategy with no distortions to attention or choices. Under unknown $\beta$, the optimal $Q$ is characterized by the implicit equation:

$$Q(\Delta v) = \frac{\Delta v}{\frac{\int \rho'_\beta(Q(\Delta v))\beta \, dF(\beta)}{\int \rho'_\beta(Q(\Delta v)) \, dF(\beta)}}, \tag{5}$$

where $\rho_\beta(x) = e^{\frac{\beta}{\lambda}x}/(1+e^{\frac{\beta}{\lambda}x})$ denotes here the logit choice probability as a function of the stakes. The optimal $Q$ scales up stakes by the inverse of a weighted average of the bias. The weights are the sensitivities $\rho'_\beta$ of the choice probabilities to the stakes. For higher material stakes, less biased types (high $\beta$) already choose correctly with high probability, so their choice probabilities are insensitive to further stake increases. The more biased types remain responsive. Thus, at higher stakes, the weighting shifts toward more biased types, who require larger emotional amplification. Put differently, mistakes in high-stakes states occur only for strongly biased agents and thus warrant larger penalties relative to the material stakes. This yields disproportionate aversion to large regrets: $Q$ is starshaped rather than linear.

We now turn to the behavioral implications of optimal regret, beginning with attention. Choice distortions are addressed below. While regret's effects on choice are well studied, its implications for attention have been overlooked. We show that regret skews attention toward extreme states.

To state this formally, we measure the *extremeness* of a state $\omega$ by the material stakes

$$|\Delta v(\omega)| := |v(a_1, \omega) - v(a_2, \omega)|.$$

We measure *attention to state* $\omega$ by the log-likelihood ratio of of the signal (equivalently, the action) conditional on that state. By equation (3), the type-$\beta$ agent's log-likelihood ratio is

$$\ell_\beta(\omega) := \log\left(\frac{p_\beta(a_1|\omega)}{p_\beta(a_2|\omega)}\right) = \frac{\beta}{\lambda} Q\big(\Delta v(\omega)\big).$$

The first-best attention strategy $p^*$ yields log-likelihood ratio $\ell^*(\omega) = \frac{1}{\lambda}\Delta v(\omega)$.

**Corollary 1.** *The ratio $\ell_\beta(\omega)/\ell^*(\omega)$ is strictly increasing in the extremeness of state $\omega$. That is, relative to the first-best, the agent pays disproportionately more attention to more extreme states.*

The intuition is as follows. The agent's attention to a state is determined by the emotion-inclusive stakes. Disproportionate aversion to large regrets—the starshapedness of $Q$—amplifies stakes more in states where material stakes are already high, skewing attention toward extreme states. Formally, the ratio $\ell_\beta(\omega)/\ell^*(\omega)$ equals $\beta Q(x)/x$ where $x = \Delta v(\omega)$, and starshapedness together with skew-symmetry imply that $Q(x)/x$ is increasing in $|x|$.

The prediction of disproportionate attention to high-stakes states is similar in spirit to models of salience or focusing based on contrast (Bordalo, Gennaioli, & Shleifer 2012; Kőszegi & Szeidl 2013). Salience effects are normally understood as a feature of automatic bottom-up attention rather than deliberate top-down attention such as rational inattention (Bordalo, Gennaioli, & Shleifer 2022). We show that such focusing effects are not only consistent with rational inattention, but are in fact optimal distortions to motivate attention.[11]

[11] Lieder, Griffiths, and Hsu (2018) offer an alternative explanation for overrepresentation of extreme events based on the resource-rationality approach.

### A.3 Behavioral Predictions for Choice under Uncertainty

This section examines the behavioral predictions of our model for choice under uncertainty. We show that the starshapedness of $Q$ delivers the empirical violations of expected utility theory associated with regret theory. At first glance, starshapedness may appear to sacrifice predictive power: convexity assumptions are central to both original regret theory and the subsequently developed generalized regret theory (Loomes & Sugden 1987). However, we find that starshaped $Q$ is sufficient. First, starshaped $Q$ implies a generalized regret representation, so our model inherits all predictions of generalized regret theory, including preference reversals of the type documented by Grether and Plott (1979). Second, in the context of original regret theory, starshapedness is equivalent to the conjunction of the Allais paradox, the common-ratio effect, and simultaneous gambling and insurance—the behavioral anomalies that motivated the convex $Q$ assumption of original regret theory. Our microfoundation thus delivers precisely the functional form needed to generate regret theory's behavioral predictions.

We first define the relevant preference representations, following Herweg and Müller (2021). Let $\mathcal{A}$ denote a set of actions.

**Definition 4.** *A preference relation $\succsim$ over $\mathcal{A}$ has an **original regret representation** if there exist $\pi \in \Delta(\Omega)$, $v : \mathcal{A} \times \Omega \to \mathbb{R}$, and a skew-symmetric, continuous function $Q : \mathbb{R} \to \mathbb{R}$ such that for all $a_i, a_j \in \mathcal{A}$,* (2) *holds.*

Our model predicts preferences with an original regret representation where $Q$ is starshaped rather than convex. Starshapedness is weaker than convexity (Bruckner & Ostrow 1962).

To compare our predictions to generalized regret theory, we introduce monetary outcomes. Let $x : \mathcal{A} \times \Omega \to \mathbb{R}$ denote the monetary outcome of action $a$ under state $\omega$. For the rest of this section, we assume the material payoff is a strictly increasing, continuous function of monetary outcomes: $v(a, \omega) = \bar{v}(x(a, \omega))$ for some strictly increasing, continuous $\bar{v} : \mathbb{R} \to \mathbb{R}$. Generalized regret theory (Loomes & Sugden 1987) works directly with monetary outcomes rather than utilities, assuming a skew-symmetric function $\Psi : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ over pairs of monetary outcomes.

**Definition 5.** *A preference relation $\succsim$ over $\mathcal{A}$ has a **generalized regret representation** if there exist $\pi \in \Delta(\Omega)$ and a skew-symmetric, continuous function $\Psi : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ such that for all $a_i, a_j \in \mathcal{A}$,*

$$a_i \succsim a_j \Leftrightarrow \sum_{\omega \in \Omega} \pi(\omega) \Psi(x(a_i, \omega), x(a_j, \omega)) \geq 0. \tag{6}$$

Original regret theory is the special case where $\Psi(x, y) = Q(\bar{v}(x) - \bar{v}(y))$. Fishburn (1989) and Sugden (1993) axiomatize generalized regret theory. Loomes and Sugden (1987) impose two additional properties on $\Psi$:[12]

- **Increasing**: for all $x, y, z \in \mathbb{R}$, $x \gtreqless y \Leftrightarrow \Psi(x, z) \gtreqless \Psi(y, z)$.

[12] Loomes and Sugden (1987) also impose an ordering property, but it is implied by increasingness and skew-symmetry.

- **Convex**: for all $x > y > z$, $\Psi(x, z) > \Psi(x, y) + \Psi(y, z)$.

Under these properties, generalized regret theory shares many predictions with original regret theory (Loomes & Sugden 1987).

**Proposition 1.** *If preferences have an original regret representation with starshaped $Q$, they have a generalized regret representation with increasing and convex $\Psi$.*

Despite the suggestive name, convexity of $\Psi$ does not require convexity of $Q$—contrary to claims in the literature. The proof is straightforward. Define $\Psi(x, y) := Q(\bar{v}(x) - \bar{v}(y))$. Then $\Psi$ inherits skew-symmetry and continuity from $Q$, so preferences have a generalized regret representation. For convexity, the inequality $\Psi(x, z) > \Psi(x, y) + \Psi(y, z)$ reduces to $Q(a + b) > Q(a) + Q(b)$ where $a := \bar{v}(x) - \bar{v}(y)$ and $b := \bar{v}(y) - \bar{v}(z)$. This is superadditivity, which follows from starshapedness.

Proposition 1 establishes that our model inherits all predictions of generalized regret theory. However, original regret theory with convex $Q$ may generate additional predictions. Does weakening convexity to starshapedness sacrifice these? We show that it does not: starshapedness is equivalent to the Allais paradox, the common-ratio effect, and simultaneous gambling and insurance—the phenomena that motivated the convexity assumption on $Q$.

The following definitions, from Loomes and Sugden (1982), formalize these phenomena. Given a prior $\pi$ over states, an action induces a lottery over monetary outcomes, allowing us to identify actions with lotteries. Two independent lotteries are identified with two actions such that the distributions over monetary outcomes induced by the prior are independent. Let $(x_1, p; x_2, 1 - p)$ denote a lottery giving monetary outcome $x_1$ with probability $p$ and $x_2$ otherwise, and let $(x, p)$ be shorthand for $(x, p; 0, 1 - p)$.

**Definition 6** (Allais Paradox)**.** *Let $X_1 = (x_1, p_1; x_2, \alpha)$ and $X_2 = (x_2, p_2 + \alpha)$ be independent lotteries where $1 \geq p_2 > p_1 > 0$ and $(1 - p_2) \geq \alpha \geq 0$. If there exists some probability $\bar{\alpha}$ such that $X_1 \sim X_2$ when $\alpha = \bar{\alpha}$, then (i) if $x_1 > x_2 > 0$, then $\alpha < \bar{\alpha} \Rightarrow X_1 \succ X_2$ and $\alpha > \bar{\alpha} \Rightarrow X_1 \prec X_2$, and (ii) if $0 > x_2 > x_1$, then $\alpha < \bar{\alpha} \Rightarrow X_1 \prec X_2$ and $\alpha > \bar{\alpha} \Rightarrow X_1 \succ X_2$.*

**Definition 7** (Common-Ratio Effect)**.** *Let $X_1 = (x_1, \lambda p)$ and $X_2 = (x_2, p)$ be independent lotteries, where $1 \geq p > 0$ and $1 > \lambda > 0$. If there exists some probability $\bar{p}$ such that $X_1 \sim X_2$ when $p = \bar{p}$, then (i) if $x_1 > x_2 > 0$, then $p < \bar{p} \Rightarrow X_1 \succ X_2$ and $p > \bar{p} \Rightarrow X_1 \prec X_2$, and (ii) if $0 > x_2 > x_1$, then $p < \bar{p} \Rightarrow X_1 \prec X_2$ and $p > \bar{p} \Rightarrow X_1 \succ X_2$.*

**Definition 8** (Simultaneous Gambling and Insurance)**.** *Let $\bar{v}$ be linear in money. Let $X_1 = (0, 1)$ and $X_2 = (x, p; -px/(1 - p), 1 - p)$ be independent lotteries where $0 < p < 1$ and $x > 0$. Then (i) for $p > 0.5$, $X_1 \succ X_2$, (ii) for $p < 0.5$, $X_1 \prec X_2$, and (iii) for $p = 0.5$, $X_1 \sim X_2$.*

We now show that these three phenomena are characterized by starshapedness.

**Proposition 2.** *Suppose preferences have an original regret representation. The following are equivalent: (i) $Q$ is starshaped; (ii) preferences exhibit the Allais paradox, the common-ratio effect, and simultaneous gambling and insurance.*

The key observation is that the arguments in Loomes and Sugden (1982) require weaker properties than convexity: the Allais paradox and common-ratio effect follow from superadditivity of $Q$, while simultaneous gambling and insurance is equivalent to starshapedness. Thus, our model predicts the behavioral anomalies that motivated original regret theory, without requiring the stronger convexity assumption.

Finally, we address a methodological point. In our model, beliefs arise endogenously from attention, whereas regret theory, following the Savage framework, takes beliefs as given. But conditional on any belief, the agent's preferences over actions in our model are exactly as in regret theory. Given the equilibrium beliefs that agents acquire, Section F shows that choices are undistorted relative to expected material utility—the agent chooses the same action as a material-utility maximizer. It is a striking result that regret solves the attention problem so effectively that its choice distortions do not materialize in equilibrium. Given other beliefs—for example, resulting from trembles in attention or exogenous information arrival—choices would be distorted exactly as regret theory predicts.

Regret theory has been applied to choices between lotteries, where the agent's belief is assumed to match the objective probabilities. We follow the standard evolutionary methodology—applicable to both biological and cultural evolution—of deriving preferences for representative environments and applying them more broadly (Robson 2001a; Rayo and Becker 2007; Netzer 2009).[13] We take the representative environment to be one where the agent can acquire information about the state. This is natural in most real-world settings, including the typical applications of regret theory: investors can research assets, patients can learn about treatments, bidders can investigate auctioned objects. Pure lottery environments—where the agent faces explicit objective probabilities—are rare outside economic experiments. Thus, we assume that when the agent faces a lottery, the preferences derived for the representative environment still apply, generating the behavioral anomalies characterized in Proposition 2.

## B More than Two Actions

Regret theory does not easily generalize to multiple actions because regret induces nontransitive preferences in pairwise choice. Several models have been proposed. Quiggin (1994) assumes that regret depends on the difference between the actual payoff and the payoff of the ex post optimal action. Loomes and Sugden (1987) propose generalized regret theory, where utility depends on the realized outcome and the outcomes of all counterfactual actions under the same state. Our solution is closest to the latter but imposes additional structure. We define a *multiaction regret solution* as follows.

**Definition 9** (Multiaction Regret Solution)**.** *Under a finite choice set $A$, a solution $(u, (p_\beta)\beta)$ to*

[13] These papers derive optimal utility functions for deterministic choice problems, then interpret the resulting preferences in risky-choice contexts—a larger leap than in our model, where the agent already faces uncertainty at the decision-making stage. More recently, Netzer, Robson, Steiner, and Kocourek (2025) extend the approach of Robson (2001a) and Netzer (2009) directly to risky choice, though they still assume preferences apply beyond the environment for which they evolved.

*the principal's problem* (P) *is a* ***multiaction regret solution*** *if there exists a function* $R$ *such that for all* $(a,\omega) \in A \times \Omega$*:*

$$u(a,\omega) = v(a,\omega) + R\big(\{v(a,\omega) - v(a',\omega)\}_{a' \in A \setminus \{a\}}\big),$$

*and for all* $a, a' \in A$*,* $\omega \in \Omega$*:*

$$v(a,\omega) > v(a',\omega) \Leftrightarrow u(a,\omega) > u(a',\omega).$$

According to a multiaction regret solution, emotions are a state-independent function of the set of payoff differences to counterfactual actions, so a permutation of counterfactual payoffs does not change the emotion, nor does shifting all payoffs in a state by a constant. Emotions preserve the strict ordering of payoffs within each state, as in Sarver (2008). Our representation imposes more structure than Loomes and Sugden (1987): utility depends on payoff differences rather than outcomes, maintaining the spirit of original regret theory. Just as original regret theory specializes generalized regret theory for two actions, our multiaction regret solution specializes the representation of Loomes and Sugden (1987) for multiple actions. Our representation is more general than Quiggin (1994): regret and rejoicing are affected by all counterfactual actions, not just the ex-post best one.

Finally, for technical reasons we assume that the principal has strict preferences over actions in each state. This can be interpreted as a genericity assumption or as the state being sufficiently fine-grained.

**Assumption 4.** *For all* $\omega \in \Omega$*, for all* $a, a' \in A$*:* $(a \neq a' \rightarrow v(a,\omega) \neq v(a',\omega))$*.*

Under these assumptions, we prove the following theorem.

**Theorem 2.** *For any finite* $A$*, the principal's problem* (P) *has a general regret solution, where* $R$ *is independent of* $A$*.*

The theorem shows that optimal emotions take the form of a multiaction regret solution. Moreover, given $\lambda$ and the distribution of $\beta$, the model can be used to compute the precise optimal emotional payoffs. While optimal emotions could depend on the whole decision problem, they reduce to a state-wise $|A|$-variable optimization problem: optimal emotions are a function of the vector of payoffs in the realized state. They are given by the socialization-cost-minimizing solution to

$$\max_{u(\cdot,\omega):\, A \to \mathbb{R}} \int \left( \lambda H\big(p_\beta(u(\omega))\big) + \sum_{a \in A} p_\beta(a|u(\omega)) v(a,\omega) \right) dF(\beta),$$

where $p_\beta(a|u(\omega))$ is given by the multinomial logit formula

$$p_\beta(a|u(\omega)) = \frac{e^{\frac{\beta}{\lambda} u(a,\omega)}}{\sum_{a' \in A} e^{\frac{\beta}{\lambda} u(a',\omega)}},$$

and $H(p_\beta(u(\omega)))$ is the entropy of the probability vector $p_\beta(u(\omega)) = (p_\beta(a|u(\omega)))_{a\in A}$. Unfortunately, the problem does not admit a simple analytical solution or comparative statics as in binary decision problems. However, once solved for a vector of payoffs, the solution applies to payoff-equivalent states in any decision problem.

## C Decision Complexity

As emphasized by Sugden (1986) and decision-justification theory (Connolly & Zeelenberg 2002), regret has a self-blame component. Taking self-blame seriously, regret theory should account for whether one could have easily known that another action would turn out better: a mistake in an easy decision is more blameworthy than the same mistake in a complex one. Thus, regret should depend on the complexity of a decision problem. Incorporating decision complexity into regret theory raises two challenges: how exactly complexity should be operationalized, and the concern that a separate regret function would be needed for each complexity level, proliferating the model's degrees of freedom. Our model addresses both challenges.

Our approach operationalizes complexity through the information cost parameter $\lambda$, which scales the cost of learning about the state, following the information-acquisition literature (e.g., Gonçalves 2024). Some decisions are inherently simpler than others: in medical decisions, the optimal treatment is easier to identify for some conditions; in auctions, the value of some goods is more easily discernible. We now let $\lambda$ vary across decision problems and index the optimal emotions accordingly, writing $R_\lambda$ and $Q_\lambda$.

The following results characterize how decision complexity affects optimal emotions in the case of two actions.

**Proposition 3.** *Under two actions:*

1. *(Scaling) For any $\lambda, \lambda' > 0$ and $x > 0$:*

$$Q_{\lambda'}(x) = \frac{\lambda'}{\lambda} Q_\lambda\left(\frac{\lambda}{\lambda'} x\right).$$

2. *(Comparative Statics) For $\lambda' < \lambda$: $Q_{\lambda'}(x) > Q_\lambda(x)$ and $|R_{\lambda'}(-x)| > |R_\lambda(-x)|$ for all $x > 0$.*

The scaling property disciplines the extension of regret theory to incorporate complexity. Measuring $Q$ at a single reference complexity suffices to determine $Q$ at any other complexity via rescaling, so no additional preference parameters are needed. The reason is that $\lambda$ enters the agent's choice probabilities only through the ratio $\beta u/\lambda$, so scaling material payoffs and complexity by the same factor scales the optimal emotion-inclusive utility by the same factor. The complexity parameter itself can be identified from state-dependent stochastic choice data (Caplin 2016): more complex problems induce choice probabilities closer to uniform.

The comparative statics show that simpler problems (lower $\lambda$) induce larger $Q_\lambda$. A larger $Q_\lambda$ means that regret and rejoicing amplify material stakes more, providing stronger incentives for attention. Thus, the incentive effects of regret are strongest in simple decision problems. This is relevant for feedback interventions, which induces regret by providing counterfactual feedback: the attentional benefits are larger in simpler problems. Mathematically, the result follows from the scaling property and starshapedness. The intuition is similar to that for starshapedness: when the right action is easy to discover, only agents with strong biases make mistakes, and these agents require larger emotional amplification to offset their bias.

Simpler problems also induce stronger regret, $|R_{\lambda'}(-x)| > |R_\lambda(-x)|$, consistent with the self-blame dimension: mistakes are punished more severely when the right action was easier to discover. Two mechanisms contribute. First, a larger $Q$ means a larger sum of regret and rejoicing. Second, a larger $Q$ increases the probability of taking the right action, so regret is experienced less often than rejoicing, making regret relatively cheaper to induce for the principal.

The next proposition characterizes the limiting behavior of optimal emotions as decision complexity becomes extreme. In both limits, emotion-inclusive utility differences become proportional to material payoff differences, so the disproportionate response to larger stakes—which drives violations of expected utility theory—disappears.

**Proposition 4** (Limits). *Under any finite action set $A$, let $(u_\lambda)_{\lambda>0}$ be a selection of solutions to the principal's problem. For all $a, a' \in A$ and $\omega \in \Omega$:*

1. *As $\lambda \to 0$: $u_\lambda(a,\omega) - u_\lambda(a',\omega) \to \big(v(a,\omega) - v(a',\omega)\big)/\underline{\beta}$.*
2. *As $\lambda \to \infty$: $u_\lambda(a,\omega) - u_\lambda(a',\omega) \to \big(v(a,\omega) - v(a',\omega)\big) \cdot \mathbb{E}[\beta]/\mathbb{E}[\beta^2]$.*

*Convergence is uniform over $v$ in any compact set.*

**Remark 1.** *Under two actions, Proposition 4 is equivalent to:*

1. *As $\lambda \to 0$: $Q_\lambda(x) \to x/\underline{\beta}$.*
2. *As $\lambda \to \infty$: $Q_\lambda(x) \to x \cdot \mathbb{E}[\beta]/\mathbb{E}[\beta^2]$.*

In very simple decision problems ($\lambda \to 0$), the optimal emotion scheme perfectly offsets the bias of the most biased type, $\underline{\beta}$. In very complex problems ($\lambda \to \infty$), the emotion scheme offsets a weighted average of biases, $\mathbb{E}[\beta^2]/\mathbb{E}[\beta]$, which places more weight on higher types. In both limits, $Q_\lambda$ is linear in stakes, so the starshapedness property that generates the Allais paradox and related anomalies vanishes. Thus, regret-based distortions to choice and attention are strongest in problems of intermediate complexity.

# IV Conclusion

Behavioral models in economics are typically developed based on explanatory power and simplicity. This paper takes a complementary approach based on the *function* of non-expected utility preferences. The usefulness of this approach is illustrated in three ways. First, it provides a criterion to arbitrate between different regret theories: our model delivers original regret theory rather than generalized regret theory, and for multiple actions, it predicts that regret depends on all counterfactual actions rather than only the ex-post best one. Second, it extends regret theory to incorporate decision complexity without proliferating degrees of freedom. Third, it generates new testable predictions regarding feedback interventions: providing counterfactual feedback increases attention but distorts it toward high-stakes states, with effects that vary systematically with decision complexity.

Several open questions remain. Our analysis focuses on symmetric decision problems for tractability. Extending the model to asymmetric problems, where some actions are ex-ante more attractive, could reveal whether optimal emotions introduce biases toward certain actions or distort decision-making to incentivize attention. For multiple actions, our model delivers a general regret solution, but whether additional properties can be obtained remains open. Finally, Appendix VI takes initial steps toward incomplete feedback, where the realized state is only partially revealed; a fuller analysis could characterize when disappointment and elation supplement regret and rejoicing.

On a broader scale, our research underscores parallels between internal psychological mechanisms, such as counterfactual comparisons, and more traditional principal-agent problems. We believe that such a perspective has the potential to profit both behavioral economics and mechanism design. Behavioral economics might glean insights from mechanism design to better understand the function and benefits of non-standard preferences. Similarly, mechanism design can turn to human psychology as a source of innovative solutions for agency problems, which evolution has already solved.

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# V Appendix A: Proofs

## A General Mechanism

This section analyzes whether the principal could benefit from using a more general mechanism than inculcating a single emotion. A general mechanism elicits a message from the agent *before* as well as *after* attention. Conditional on the messages, an action recommendation is sent and an emotional utility is induced conditional on the messages and the outcome. By the revelation principle it is sufficient to restrict attention to mechanisms, where the first message corresponds to the agent's type. We maintain the assumption that any emotion induces an symmetric decision problem for the agent. First, we analyze the case when the mechanism elicits a message only before attention. At the end of the proof, we argue that eliciting a message after attention can be reduced to a mechanism of the former case.

The principal chooses a vector of emotions $m_\beta \colon A \times \Omega \to \mathbb{R}$ and attention strategies $p_\beta \colon \Omega \to \Delta(A)$, indexed by the agent's type, subject to incentive compatibility and symmetry. The incentive-compatibility constraint below captures both truthful reporting of one's type as well as the subsequent attention strategy, accounting for the possibility of double deviations. The principal's problem (P') is

$$\max_{(u_\beta, p_\beta)_\beta} \int \left( -c(p_\beta) + \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega) p_\beta(a|\omega) v(a,\omega) \right) dF(\beta) \tag{P'}$$

$$\text{s.t.}$$

$$\forall \beta \colon \ (\beta, p_\beta) \in \arg\max_{\hat\beta, \hat p} -c(\hat p) + \beta \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega)\hat p(a|\omega) u_{\hat\beta}(a,\omega) \tag{IC'}$$

$$\forall \beta \colon \ u_\beta \text{ is symmetric.} \tag{SYM'}$$

Out of the solutions $(m_\beta, p_\beta)_\beta$, the principal chooses the one that minimizes the implementation cost of emotions,

$$\int \left( \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega) p_\beta(a|\omega) \phi\big(m_\beta(a,\omega)\big) \right) dF(\beta).$$

**Lemma 1.** *Let* $|\Omega| = 2$*. Under Assumptions 1 to 3, for any* $(m_\beta, p_\beta)_\beta$ *satisfying* (IC') *and* (SYM')*, there exists* $(m'_\beta, p'_\beta)_\beta$ *with* $m'_\beta = m'_{\beta'}$ *for all* $\beta, \beta'$*, satisfying* (IC') *and* (SYM')*, that achieves a weakly higher payoff for the principal in* (P')*. Thus, the formulation* (P) *is without loss.*

The proof of Lemma 1 shows that offering a menu makes types sort into emotions to the principal's disadvantage. Ideally, the principal would want to offer stronger incentives to types with lower $\beta$ than to types with higher $\beta$ because the former group is more biased. However, higher types have a higher return to attention and pick stronger incentives than lower types, contrary to the principal's aim. This leads to higher types paying more attention and lower types less than under a single emotion scheme. Hence, offering a menu increases the variance of attention, decreasing the principal's utility. Our proof does not rely on the specific shape of mutual information and we show the proposition for any posterior-separable attention cost (see section VI for a definition) with strictly concave and differentiable $H$ that is *symmetric in the states*.

Our result contrasts with related findings in the literature. Osband (1989) shows that in their model offering a menu of emotion schemes can in principle benefit the principal, but using numerical simulations he shows that these gains are very small. In contrast to Osband, in our setup, heterogeneity concerns the agent's bias instead of his cost parameter. Thus, the first-best attention strategy is the same for every agent, which the proof relies on. In Yoder (2022), the principal can effectively screen agents because the acquired posterior is contractible. In contrast, under hidden attention, the principal can induce only gross value functions that are convex in the posterior. Offering a menu makes the gross value function more convex, leading to further distortions at the top and bottom.

*Proof.* Under symmetry of $v$, if there are two states, $\Omega = \{\omega_1, \omega_2\}$, then one of the following needs to be the case. The payoff $v$ is state-wise constant, or $\mu(\omega_1) = \mu(\omega_2) = \frac{1}{2}$ and there are two actions, $A = \{a_1, a_2\}$, with symmetric payoffs, $v(a_1, \omega_1) = v(a_2, \omega_2)$ and $v(a_1, \omega_2) = v(a_2, \omega_1)$. If there were more than two actions and $v$ is not state-wise constant, then more than two states are necessary to have all permutations of $(v(a, \omega))_{a \in A}$ be equiprobable.

If the payoffs are state-wise constant, then the optimal attention strategy is to pay zero attention, which is achieved at the smallest implementation cost by setting the emotions to 0 for all types. Thus, all $u_\beta$ are the same.

Consider, then, the other case that or $\mu(\omega_1) = \mu(\omega_2) = \frac{1}{2}$ and there are two actions, $A = \{a_1, a_2\}$, with symmetric payoffs, $v(a_1, \omega_1) = v(a_2, \omega_2)$ and $v(a_1, \omega_2) = v(a_2, \omega_1)$. Without loss, the payoff matrix for the principal can be written as

| | $\omega_1$ | $\omega_2$ |
|---|---|---|
| $a_1$ | $v$ | $0$ |
| $a_2$ | $0$ | $v$ |

where $v > 0$.

**Agent's Behavior** Turning to the agent, we first analyze the agent's behavior given some $u\colon A \times \Omega \to \mathbb{R}$ in the menu of utility functions $(u_\beta)_\beta = (v + m_\beta)_\beta$ that the principal offers.[14] By the

[14] By a change of variables, the principal chooses $u$ directly instead of $m$.

symmetry and an argument analogous to the above, the payoff matrix of any such $u$,

| | $\omega_1$ | $\omega_2$ |
|---|---|---|
| $a_1$ | $u(a_1, \omega_1)$ | $u(a_1, \omega_2)$ |
| $a_2$ | $u(a_2, \omega_1)$ | $u(a_2, \omega_2)$ |

must satisfy one of the following.

Either $u$ is state-wise constant, that is $u(a_1, \omega_1) = u(a_2, \omega_1)$ and $u(a_2, \omega_1) = u(a_1, \omega_1)$, or $u(a_1, \omega_1) = u(a_2, \omega_2)$ and $u(a_1, \omega_2) = u(a_2, \omega_1)$. In the first case, we can make $u$ constant over all outcomes without changing the agent's behavior (he pays no attention in either case), so it is captured by the second case, which we analyze below. We can write the agent's payoff matrix as

| | $\omega_1$ | $\omega_2$ |
|---|---|---|
| $a_1$ | $\underline{u} + \Delta u$ | $\underline{u}$ |
| $a_2$ | $\underline{u}$ | $\underline{u} + \Delta u$ |

where $\underline{u} = u(a_1, \omega_2) = u(a_2, \omega_1)$ and $\Delta u = u(a_1, \omega_1) - u(a_2, \omega_1) = u(a_2, \omega_2) - u(a_1, \omega_2)$.

We characterize the optimal attention strategy given some chosen utility function $u$. Following Caplin, Dean, and Leahy (2019), we solve for the optimal distribution over posteriors, induced by the attention strategy, analogous to Bayesian persuasion models. Under two states, we can identify posteriors $\gamma \in \Delta(\Omega)$ with the probability in $[0, 1]$ they assign to state $\omega_1$, which makes this posterior approach especially attractive. The problem of agent $\beta$ given $u$ is to pick a Bayes-consistent distribution over posteriors to maximize the expectation of the *net utility* function $N_\beta$ (Caplin, Dean, and Leahy 2019):

$$\max_{\tau \in \Delta[0,1]} \mathbb{E}_\tau [N_\beta(\gamma | u)]$$
$$\text{s.t.}$$
$$\mathbb{E}_\tau[\gamma] = \tfrac{1}{2}$$

Here, $N_\beta(\gamma|u)$ is the net utility of posterior $\gamma$ for the agent of type $\beta$ given $u$, which is

$$N_\beta(\gamma|u) = \beta(\underline{u} + \max\{\gamma \Delta u, (1-\gamma)\Delta u\}) + \lambda H(\gamma).$$

The first summand of the net utility $N_\beta(\cdot|u)$ captures the *gross value*, or instrumental value, of attention and the second the attention cost. The gross value is continuous, convex, symmetric around $1/2$, and piecewise linear as the maximum of linear functions. The attention cost is mutual information, $H(\gamma) = \gamma \log(\gamma) + (1-\gamma)\log(1-\gamma)$, which is strictly concave, differentiable and symmetric around $1/2$. These are the only properties of $H$ that we make use of for the proof.

As known from the literature on Bayesian persuasion, the solution can be obtained by concavifying the net utility function $N_\beta(\cdot|u)$. As we show next, this Bayesian persuasion problem takes a particularly simple form due to symmetry.

The net utility function has one global maximum if $\Delta u = 0$ and two global maxima otherwise. Restricted to $[1/2, 1]$, the net utility function attains its maximum by continuity. On $[1/2, 1]$, the net utility function is the sum of the linear gross value and the strictly concave attention cost, so the maximum must be unique. By symmetry and differentiability of $H$, we have $H'(1/2) = 0$, so the maximum is at $1/2$ only if $\Delta u = 0$. In fact, by differentiability the maximum is characterized by $-\lambda H'(\gamma) = \beta \Delta u$ if $-\lambda H'(1) > \beta \Delta u$ and $\gamma = 1$ otherwise. By symmetry of $H$ around $1/2$, the net utility function $N_\beta(\cdot|u)$ is symmetric around $1/2$, so the net utility function attains another maximum at the reflection point around $1/2$.

Because the maxima are symmetric around the prior probability $1/2$, there is a unique Bayes-consistent $\tau$ whose support is the set of global maxima. This is the unique $\tau$ that attains the highest expected posterior net utility because each posterior in its support maximizes the net utility function. This $\tau$ is either degenerate at $1/2$ or binary and symmetric around $1/2$. The agent selects action $a_1$ when the realized posterior is greater than $1/2$ and $a_2$ when the realized posterior is less than $1/2$, if $\Delta u > 0$ and vice versa if $\Delta u < 0$. Thus, given $v > 0$, the principal never offers an emotion-inclusive utility with $\Delta u < 0$, which would induce the agent to pay costly attention only to select the wrong action.

Thus, while the agent chooses a high-dimensional distribution over posteriors, we can characterize his choice by the one-dimensional maximizer of the net utility function on $[1/2, 1]$, which corresponds to the higher posterior the agent acquires. As written above, the maximizer is characterized by $\lambda - H'(\gamma) = \beta \Delta u$ if $-\lambda H'(1) > \beta \Delta u$ and $\gamma = 1$ otherwise. Because $H$ is strictly concave and symmetric, $-H'$ is strictly increasing on $[1/2, 1]$ with $-H'(1/2) = 0$. Thus, the maximum is strictly increasing in $\Delta u$ as long as the optimal $\gamma$ is less than 1 and constant afterwards.

The complete strategy of the agent consists of choosing an emotion scheme and subsequently an attention strategy. By the above, the agent's behavior can be described as choosing a net utility function and subsequently the maximum of the net utility function on $[1/2, 1]$. This is equivalent to choosing the maximum on $[1/2, 1]$ of the pointwise maximum $N_\beta(\cdot|(u_\beta)_\beta)$ of the collection of net utility functions,

$$\begin{aligned} N_\beta(\gamma|(u_\beta)_\beta) =& \max_{\hat{\beta}} \{N_\beta(\gamma, u_{\hat{\beta}})\} \\ =& \beta \max_{\hat{\beta}} \{\underline{u}_{\hat{\beta}} + \gamma \Delta u_{\hat{\beta}}\} + \lambda H(\gamma), \text{ for } \gamma \geq 1/2. \end{aligned}$$

We call $N_\beta(\gamma|(u_\beta)_\beta)$ the *effective* value function. By $\Delta u_\beta > 0$ for all $\beta$, the effective value function is supermodular on $[1/2, 1]$ in $\beta$ and $\gamma$ and thus the optimal $\gamma$ is non-decreasing in $\beta$. The effective value function on $[1/2, 1]$ is the sum of the maximum of increasing, linear functions and a concave attention cost. The maximum of increasing, linear functions is increasing and convex. Thus, non-decreasing $\gamma$ implies a non-decreasing choice of the slope $\Delta u_{\hat{\beta}}$.

To summarize, for any type of the agent, the menu of emotion-inclusive utility functions induces an effective value function and the type's acquired signal structure can be described as picking the posterior in $[1/2, 1]$ that maximizes this net utility function. We have shown that (1) a higher $\beta$ implies a weakly more extreme posterior, and (2) a weakly higher slope $\Delta u_\beta$ of the gross value function. Given the choice of an emotion-inclusive utility, (3) the agent's posterior is a strictly increasing function of the slope of the gross value function, if the posterior is below 1.

**Principal's Problem** Turning back to the principal, her net utility function is

$$N_P(\gamma) = \max\{\gamma v, (1-\gamma)v\} + \lambda H(\gamma).$$

Her value from the symmetric distribution with support $\{\gamma, 1-\gamma\}$ is just $N_P(\gamma)$ by symmetry of $N_P$. The principal's value function $N_P$ is strictly concave on $[1/2, 1]$ because it is the sum of the linear gross value and the strictly concave attention cost. Strict concavity implies strict quasi-concavity, which we are using below.

Suppose for the sake of contradiction that there are types $\beta, \beta'$ such that $u_\beta \neq u_{\beta'}$. We show that the principal strictly benefits from offering only the emotion-inclusive utility $u_{\beta^*}$ of a type $\beta^*$ that selects the most favorable posterior to the principal,

$$\beta^* \in \underset{\beta \in \text{supp}(\beta)}{\arg\max}\, N_P(\gamma_\beta).$$

Such a type $\beta^*$ exists by $\text{supp}(\beta)$ being finite.

By (2), greater types than $\beta^*$ select a weakly higher slope and smaller types a weakly smaller slope of the gross value function. Offering only the emotion-inclusive utility $u_{\beta^*}$, by (3) weakly increases the posterior for type less than $\beta^*$, weakly decreases the posterior for types greater than $\beta^*$, and does not change the posterior of $\beta^*$. Because the ordering of posteriors, (1), still holds due to supermodularity of the new value function, each posterior moves weakly towards $\gamma_{\beta^*}$. By strict quasi-concavity of the principal's value function on $[1/2, 1]$ and $\gamma_{\beta^*}$ yielding a weakly higher value to the principal, the principal benefits weakly from the induced changes and benefits strictly from strict changes. Thus, the principal is weakly better off offering only one emotion scheme.

It remains to show that in case of indifferences, the principal cannot decrease the implementation cost by offering different emotion schemes. By the above, the principal is strictly better off if some posterior moved strictly. The only case that no posterior changed, is when the only types whose emotion-inclusive utility changed selects $\gamma = 1$ at their original and at the new emotion-inclusive utility. By construction, the new value function is below the original induced value function at $\gamma = 1$, so it entails a smaller emotional payoff after selecting the better action under the realized state. Because $\beta \leq 1$, optimality entails that the slope of the constructed gross value function is at least $v$. Then, the implementation cost being minimized entails positive emotions at selecting the better action under the realized state. Thus, the new emotion-inclusive utility assigning smaller emotions implies a reduction of the implementation cost.

**General Mechanism** As mentioned above, the principal could use an even more general mechanism. That is, she could elicit an additional message from the agent after attention and condition the emotion scheme on both the message before and after attention. We maintain the restriction that any assigned emotion scheme $m$ is such that $v + m$ is symmetric. Here, we sketch how the proof can be extended to account for such mechanisms. We show that by starting with a general mechanism, we can construct a mechanism *without* a message after attention that is equally good for the principal. Then, the above proof can be applied. The argument below uses that if the agent acquires symmetric posteriors, there is no difference between eliciting a message before and after attention, as we have shown above for symmetric value functions. Using this, we show that the principal might as well offer only one emotion scheme given the first message from the agent (thus making the second message meaningless). The argument proceeds in three steps.

First, given an incentive-compatible general mechanism, fix an arbitrary type $\beta$ of the agent and consider the agent's attention problem following the first message. We can, again, by direct revelation identify the first message with the agent's type, which the agent truthfully reveals. The agent's attention problem still consists of concavifying a symmetric value function. That is because analogous to the argument above, the agent faces an effective value function, denoted by $\hat{N}_\beta$, given by the maximum of the value functions induced by the emotion schemes, which the agent can choose from after attention. Because each emotion scheme induces a symmetric value function, the maximum of the value functions is symmetric, too. This implies that the set of maximizers $M_\beta$ of the value function $\hat{N}_\beta$ is symmetric around $1/2$. Any Bayes-consistent distribution with support at only the maximizers of the $\hat{N}_\beta$ is a solution to the attention problem of type $\beta$.

Second, there is a symmetric, principal's preferred solution to the attention problem of type $\beta$. The principal's value function is symmetric around $1/2$, too. Hence, the principal prefers, out of the solutions to the agent's problem, those whose support includes only posteriors that give the highest value to the principal. Formally, let $M_{P,\beta} := \arg\max_{\gamma \in M_\beta} N_P(\gamma)$, which is symmetric around $1/2$. The principal's preferred solutions to the attention problem of type $\beta$ are the Bayes-consistent distributions with support in $M_{P,\beta}$. Because $M_{P,\beta}$ is symmetric and the prior is at $1/2$, there is a Bayes-consistent, binary, symmetric distribution with support in $M_{P,\beta}$.

Third, starting with a mechanism that elicits a message before and after attention, we can construct an equally good mechanism that elicits a message only before attention. As we are allowing the principal to pick her preferred solution when the agent is indifferent, for any type we can select a principal's preferred solution to his attention problem that is moreover symmetric and binary by the above. By symmetry, this solution makes any agent pick the same emotion scheme (or, send the same second message) after each acquired posterior. This solution can also be implemented by offering to every agent *only this one emotion scheme*, after the first message. Offering less choice to the agent does not not violate any other incentive-compatibility constraints. Finally, we can apply the proof above to the resulting mechanism. □

## B Theorem 2

*Proof.* Let $A = \{a_1, \dots, a_n\}$. Because the principal chooses $u$ that is symmetric by (SYM), the (IC) constraint implies that the state-dependent choice probabilities are (Matějka & McKay 2015):

$$p_\beta(a|\omega) = \frac{e^{\frac{\beta}{\lambda}u(a,\omega)}}{\sum_{a'\in A} e^{\frac{\beta}{\lambda}u(a',\omega)}} = \frac{e^{\frac{\beta}{\lambda}(u(a,\omega)-u(a_1,\omega))}}{1+\sum_{a'\neq a_1} e^{\frac{\beta}{\lambda}(u(a',\omega)-u(a_1,\omega))}}. \tag{7}$$

By (7), the choice probabilities depend only on state-wise utility differences with respect to some reference action. Using this, we show that the principal's problem separates by state.

Substituting (7) into (P), the principal chooses only $u$ and the (IC) constraint is satisfied by construction. The resulting problem consists of an outer minimization problem regarding the socialization cost $\phi(u(a,\omega) - v(a,\omega))$ and an inner problem to maximize the material utility net of attention cost, subject to the symmetry constraint:

$$\min_{u\colon A\times\Omega\to\mathbb{R}} \int \left( \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega) p_\beta(a|\omega)\phi\big(u(a,\omega) - v(a,\omega)\big)\right) dF(\beta) \tag{8}$$

s.t.

$$u \in \underset{\substack{u\colon A\times\Omega\to\mathbb{R},\\ u \text{ is symmetric}}}{\arg\max} \int \sum_{\omega\in\Omega} \mu(\omega) \left( \lambda H(p_\beta(\omega)) + \sum_{a\in A} p_\beta(a|\omega) v(a,\omega)\right) dF(\beta) \tag{9}$$

where $p_\beta(a|\omega)$ is given by (7) and $H(p_\beta(\omega)) = \sum_{a\in A} -p_\beta(a|\omega)\log(p_\beta(a|\omega))$ is the entropy of the conditional choice probabilities. We have used that under symmetric payoffs, the unconditional choice probabilities are all equal to $1/n$, so the constant entropy of the unconditional choice probabilities can be dropped.

**Inner Maximization Problem.** We first relax the inner maximization problem by dropping the symmetry constraint. Then, the inner objective is separable by state as $p_\beta(\omega)$ depends only on the vector of state-wise utilities, $u(\omega) := (u(a_1,\omega), \dots, u(a_n,\omega))$. Below, we show that any solution to the relaxed state-wise maximization problem

$$\max_{u(\cdot,\omega)\colon A\to\mathbb{R}} \int \left( \lambda H(p_\beta(\omega)) + \sum_{a\in A} p_\beta(a|\omega) v(a,\omega)\right) dF(\beta) \tag{10}$$

preserves the strict order of utilities and that a solution exists. Then, we show that there is a symmetric utility function that maximizes (10) state-by-state. By every state having positive probability, this implies that any solution to the constrained inner-maximization problem solves the relaxed state-wise problem (10) in every state.

**Lemma 2.** *Any solution to the relaxed state-wise maximization problem (10) satisfies*

$$\forall a, a' \in A\colon v(a,\omega) > v(a',\omega) \Rightarrow u(a,\omega) > u(a',\omega).$$

*Proof.* If $v(a,\omega) > v(a',\omega)$ but $u(a,\omega) < u(a',\omega)$, exchanging $u(a,\omega)$ and $u(a',\omega)$ leaves entropy unchanged (by symmetry of $H$ in the probabilities) while improving expected material utility. Hence $u(a,\omega) \geq u(a',\omega)$ at any maximum.

For strict inequality, suppose $v(a,\omega) > v(a',\omega)$ and $u(a,\omega) = u(a',\omega)$. Setting $\tilde{u}(a,\omega) = u(a,\omega) + \varepsilon$ and $\tilde{u}(a',\omega) = u(a',\omega) - \varepsilon$: by symmetry of (7) at $\varepsilon = 0$, we have $p_\beta(a|\omega) = p_\beta(a'|\omega)$, and by symmetry of $H$, $\frac{dH}{d\varepsilon} = 0$, while $\frac{d}{d\varepsilon}\sum_{\hat{a}} p_\beta(\hat{a}|\omega)v(\hat{a},\omega) > 0$. Hence $u(a,\omega) > u(a',\omega)$. $\square$

By the genericity Assumption 4, the lemma implies

$$v(a,\omega) > v(a',\omega) \Leftrightarrow u(a,\omega) > u(a',\omega).$$

Before we show existence, we rewrite the state-wise objective in (10), which we denote by $G(u(\omega))$, as

$$G(u(\omega)) = \int \left( \sum_{a\in A} p_\beta(a|u(\omega))\big(v(a,\omega) - \beta u(a,\omega)\big) + \lambda \log\left(\sum_{a\in A} e^{\frac{\beta}{\lambda}u(a,\omega)}\right)\right) dF(\beta). \tag{11}$$

The first term is the difference between the principal's and the agent's utility and the second term is the agent's state-wise utility by Caplin, Dean, and Leahy (2019).

It will be useful to know the derivatives of $G$ and $p_\beta(a|\omega)$ with respect to the components of $u$:

$$\frac{dG(u(\omega))}{du(a,\omega)} = \int \left(\sum_{a'\in A} \frac{dp_\beta(a'|\omega)}{du(a,\omega)}\big(v(a',\omega) - \beta u(a',\omega)\big)\right) dF(\beta) \tag{12}$$

$$\frac{dp_\beta(a|\omega)}{du(a,\omega)} = \frac{\beta}{\lambda} p_\beta(a|\omega)(1 - p_\beta(a|\omega)) \tag{13}$$

$$\frac{dp_\beta(a'|\omega)}{du(a,\omega)} = -\frac{\beta}{\lambda} p_\beta(a'|\omega)p_\beta(a|\omega), \text{ for } a \neq a'. \tag{14}$$

**Lemma 3.** *The relaxed state-wise maximization problem (10) has a solution. Moreover, any solution satisfies*

$$u(a_i,\omega) \leq (i-1)\frac{v(a_n,\omega)}{\inf \beta}$$

*for $i = 1,\ldots,n$, assuming $v(a_1,\omega) \leq \cdots \leq v(a_n,\omega)$ and normalizing $u(a_1,\omega) = v(a_1,\omega) = 0$.*

*Proof.* We show that any $u(\cdot,\omega) \in \mathbb{R}^A$ is weakly dominated by some $\tilde{u}(\cdot,\omega) \in \mathbb{R}^A$ in a compact set, on which the maximum is obtained by continuity of the objective.

The normalizations $v(a_1,\omega) = u(a_1,\omega) = 0$ are without loss: shifting all $u(a,\omega)$ by a constant does not affect choice probabilities, and shifting all $v(a,\omega)$ by a constant $c$ changes the objective by exactly $c$.

By Lemma 2, if $v(a,\omega) > v(a',\omega)$ and $u(a',\omega) > u(a,\omega)$, exchanging $u(a',\omega)$ and $u(a,\omega)$ improves the objective; if $v(a,\omega) = v(a',\omega)$ and $u(a',\omega) > u(a,\omega)$, exchanging leaves the objective unchanged. Thus, we can restrict attention to $0 = u(a_1,\omega) \leq \ldots \leq u(a_n,\omega)$.

For the upper bound, suppose for some $m \in \{2, \dots, n\}$ that $u(a_m, \omega) > u(a_{m-1}, \omega) + \frac{v(a_n,\omega)}{\inf \beta}$. Then for all $i \geq m$, $k \leq m-1$:

$$u(a_i, \omega) \geq u(a_m, \omega) > u(a_{m-1}, \omega) + \frac{v(a_n, \omega)}{\inf \beta} \geq u(a_k, \omega) + \frac{v(a_n, \omega)}{\inf \beta}. \tag{15}$$

Denote $W_k := v(a_k, \omega) - \beta u(a_k, \omega)$. Using (12)–(14), the derivative of decreasing $u(a_m, \omega), \dots, u(a_n, \omega)$ uniformly is

$$\begin{aligned}
-\sum_{i=m}^{n} \frac{dG(u)}{du(a_i, \omega)} &= \sum_{i=m}^{n} \int \frac{\beta}{\lambda} p_\beta(a_i|\omega) \left( \sum_{k=1, k\neq i}^{n} p_\beta(a_k|\omega) \Big[ W_k - W_i \Big] \right) dF(\beta) \\
&= \sum_{i=m}^{n} \int \frac{\beta}{\lambda} p_\beta(a_i|\omega) \left( \sum_{k=1}^{m-1} p_\beta(a_k|\omega) \Big[ W_k - W_i \Big] \right) dF(\beta) \\
&\geq \sum_{i=m}^{n} \int \frac{\beta}{\lambda} p_\beta(a_i|\omega) \left( \sum_{k=1}^{m-1} p_\beta(a_k|\omega) \Big[ \beta\big(u(a_i, \omega) - u(a_k, \omega)\big) - v(a_n, \omega) \Big] \right) dF(\beta),
\end{aligned}$$

which is strictly positive by (15). Hence $u(a_i, \omega) \leq (i-1)\frac{v(a_n,\omega)}{\inf \beta}$, which defines a compact set. $\square$

**Lemma 4.** *There exists a symmetric utility function $u \colon A \times \Omega \to \mathbb{R}$ that maximizes (10) state-by-state.*

*Proof.* The objective (10) is symmetric under permutation of actions: permuting both $v(\cdot, \omega)$ and $u(\cdot, \omega)$ by the same permutation leaves the objective unchanged. Thus, if $u(\cdot, \omega)$ solves (10) given $v(\cdot, \omega)$, then permuting $v(\cdot, \omega)$ leads to the same permutation of the solution candidates. Define an equivalence relation on $\Omega$: two states are equivalent if their utility-vectors $v(\cdot, \omega)$ are a permutation of each other. For each equivalence class, pick a representative state and a solution $u(\cdot, \omega)$. For other states $\omega'$ in the equivalence class, assign the analogous permutation of $u(\cdot, \omega)$, which is a solution given $v(\cdot, \omega')$. This procedure is well-defined because by Assumption 4 and Lemma 2, all components of $v(\cdot, \omega)$ and $u(\cdot, \omega)$ are pairwise different, so any permutation is uniquely determined. By symmetry of $v$, the resulting $u$ is symmetric. $\square$

We have thus shown that the inner maximization problem separates into state-wise maximization of (10). Next, we show that the overall principal's problem separates by state.

**Outer Minimization Problem.** Because $p_\beta(a|u(\omega))$ depends only on the state-wise utility vector $u(\omega)$ by (7), the outer objective (8) separates by state. Moreover, if a solution to the principal's problem exists, then by symmetry of $v$ and an argument analogous to Lemma 4—using that (8) is symmetric under permutation of actions—a symmetric solution exists. We can therefore drop the symmetry constraint, and the principal's problem separates into state-wise problems. It follows that any solution must solve for every state $\omega \in \Omega$ the *state-wise principal's problem* (16):

$$\begin{aligned} \min_{u(\cdot,\omega)\colon A\to\mathbb{R}} \int \left( \sum_{a\in A} p_\beta(a|u(\omega))\phi\big(u(a,\omega) - v(a,\omega)\big) \right) dF(\beta) \\ \text{s.t.} \quad u(\cdot,\omega) \in \operatorname*{arg\,max}_{u(\cdot,\omega)\colon A\to\mathbb{R}} G(u(\omega)). \end{aligned} \tag{16}$$

**Lemma 5.** *A solution to the state-wise principal's problem (16) exists. The solution is unique if the solution to the inner maximization problem is unique up to a common additive constant.*

*Proof.* Reparametrize $(u(a_1,\omega),\dots,u(a_n,\omega))$ by $(u_1, \Delta u_2, \dots, \Delta u_n)$ where $\Delta u_i = u(a_i,\omega) - u(a_1,\omega)$ and $u_1 = u(a_1,\omega)$. The choice probabilities $p_\beta(a_i|u(\omega))$ depend only on $(\Delta u_2,\dots,\Delta u_n)$, so the inner maximization problem (9) constrains only the $\Delta u_i$'s, leaving $u_1$ unrestricted. The outer objective becomes

$$\sum_{i=1}^{n} \phi\big(u_1 + \Delta u_i - v(a_i,\omega)\big) \int p_\beta(a_i|u(\omega)) dF(\beta).$$

By assumption, $\phi$ is strictly convex and minimized at 0. Thus, for fixed $(\Delta u_2,\dots,\Delta u_n)$, the objective is strictly convex and radially unbounded in $u_1$. Since $\Delta u_i$ is bounded by Lemma 3, this implies a bound on $u_1$. By Berge's maximum theorem, the set of maximizers from the inner problem is compact. Hence the minimum is attained by continuity.

For uniqueness: if $(\Delta u_2,\dots,\Delta u_n)$ is unique from the inner problem, then $u_1$ is unique by strict convexity. □

For $c \in \mathbb{R}$, let $\mathbf{c} := (c,\dots,c) \in \mathbb{R}^A$.

**Lemma 6.** *$u(\cdot,\omega) \in \mathbb{R}^A$ solves the state-wise principal's problem (16) given $v(\cdot,\omega) \in \mathbb{R}^A$ if and only if $u(\cdot,\omega) + \mathbf{c}$ does so given $v(\cdot,\omega) + \mathbf{c}$.*

*Proof.* The constraint set (argmax of the inner problem) is unchanged by shifting $v(\cdot,\omega)$ by $\mathbf{c}$, since this shifts the principal's utility by $c$ independent of the agent's choice. The constraint set is also closed under shifting $u(\cdot,\omega)$ by $\mathbf{c}$, since this does not affect choice probabilities. Thus, if $u(\cdot,\omega)$ is in the constraint set given $v(\cdot,\omega)$, then $u(\cdot,\omega) + \mathbf{c}$ is in the constraint set given $v(\cdot,\omega) + \mathbf{c}$.

The outer objective is unchanged if $\mathbf{c}$ is added to both $u(\cdot,\omega)$ and $v(\cdot,\omega)$, since choice probabilities and $\phi(u(a_i,\omega) - v(a_i,\omega))$ are unaffected. Combined with the above, if $u(\cdot,\omega)$ minimizes the socialization cost given $v(\cdot,\omega)$, then $u(\cdot,\omega) + \mathbf{c}$ minimizes it given $v(\cdot,\omega) + \mathbf{c}$. The converse follows from shifting by $-\mathbf{c}$. □

**Lemma 7.** *There exists a solution $u$ to the principal's problem (8) and a function $R$, such that for all $(a,\omega) \in A \times \Omega$,*

$$u(a,\omega) = v(a,\omega) + R(\{v(a,\omega) - v(a',\omega)\}_{a'\in A\setminus\{a\}}).$$

*Proof.* Define a partition $\mathbb{P}$ on the state space $\Omega$ based on the equivalence relation that the utility vectors $v(\cdot,\omega)$ of two states are the same up to permutations, shifts of all components by a constant, or combinations of both. For each partition element $P \in \mathbb{P}$, pick a representative state $\omega \in P$ and define $u(\cdot, \omega)$ to be a solution to the state-wise principal's problem, which exists by Lemma 5.

For each other state $\omega' \in P$, there exist a unique $c \in \mathbb{R}$ and a unique permutation $\tilde{v}(\cdot, \omega)$ of $v(\cdot, \omega)$ such that $v(\cdot, \omega') = \mathbf{c} + \tilde{v}$ (uniqueness follows from Assumption 4). Define $u(\cdot, \omega')$ to be the analogous permutation of $u(\cdot, \omega)$ plus $\mathbf{c}$. This gives a solution to the state-wise principal's problem under $\omega'$ by symmetry and Lemma 6. By symmetry of $v$, the constructed $u$ is symmetric and solves the principal's problem.

By construction, $u(\cdot, \omega)$ is a function only of the material utility vector $v(\cdot, \omega)$. Denote this function by $M \colon \mathbb{R}^A \to \mathbb{R}^n$ and by $M_i$ its $i$-th component. By construction, permutations of the arguments of $M$ permute the image analogously, that is, $M$ commutes with permutations. Define

$$R(\{v_1, \ldots, v_{n-1}\}) := M_n(-v_1, \ldots, -v_{n-1}, 0),$$

which is well-defined because $M_n$ is invariant under permutations of $v_1$ through $v_{n-1}$. By Lemma 6 and $M$ commuting with permutations:

$$\begin{aligned}
u(a_i, \omega) &= M_i(v(a_1, \omega), \ldots, v(a_n, \omega)) \\
&= v(a_i, \omega) + M_i(v(a_1, \omega) - v(a_i, \omega), \ldots, 0, \ldots, v(a_n, \omega) - v(a_i, \omega)) \\
&= v(a_i, \omega) + M_n(v(a_1, \omega) - v(a_i, \omega), \ldots, v(a_n, \omega) - v(a_i, \omega), 0) \\
&= v(a_i, \omega) + R(\{v(a_i, \omega) - v(a', \omega)\}_{a' \in A \setminus \{a_i\}}).
\end{aligned} \qquad \square$$

Note that the constructed solution depends only on the state-wise material utilities as well as the distribution over $\beta$ and $\lambda$. Thus, the $R$ is independent of the choice set $A$. $\square$

## C Theorem 1

*Proof.* By Theorem 2, there exists a solution with

$$u(a, \omega) = v(a, \omega) + R(v(a, \omega) - v(a_-, \omega)), \tag{17}$$

provided $v(a, \omega) \neq v(a_-, \omega)$ for all states. We first extend this to states where $v(a_1, \omega) = v(a_2, \omega)$, then establish uniqueness of $Q$ and $R$, and finally verify Properties 1–2 and continuity.

**Extension to $v(a_1, \omega) = v(a_2, \omega)$.** If $v(a_1, \omega) = v(a_2, \omega)$, the expected material utility in the state-wise principal's problem (16) is unaffected by choice probabilities. Entropy is maximized when $p_\beta(a_1|\omega) = p_\beta(a_2|\omega) = 1/2$, achieved iff $u(a_1, \omega) = u(a_2, \omega)$. Since the socialization cost is minimized at 0, the unique solution is $u(a_1, \omega) = u(a_2, \omega) = v(a_1, \omega) = v(a_2, \omega)$. This is consistent with (17) and $R(0) = 0$, and $u = v$ maintains symmetry.

**First-order condition.** Fix a state $\omega \in \Omega$ and define

$$\begin{aligned}\Delta u &:= u(a_2, \omega) - u(a_1, \omega)\\ \Delta v &:= v(a_2, \omega) - v(a_1, \omega)\\ \rho_\beta(x) &:= \frac{e^{\frac{\beta}{\lambda}x}}{1 + e^{\frac{\beta}{\lambda}x}}.\end{aligned}$$

We have $p_\beta(a_2|\omega) = \rho_\beta(\Delta u)$. Using $p_\beta(a_1|\omega) = 1 - p_\beta(a_2|\omega)$, the first-order condition (12) becomes

$$\int \rho'_\beta(\Delta u)\big(\Delta v - \beta \Delta u\big)\, dF(\beta) = 0, \tag{18}$$

which can be rewritten as

$$\Delta u = \Delta v \left( \frac{\int \rho'_\beta(\Delta u)\beta\, dF(\beta)}{\int \rho'_\beta(\Delta u)\, dF(\beta)} \right)^{-1}. \tag{19}$$

The term in parentheses is a weighted average of $\beta$, with weights proportional to $\rho'_\beta(\Delta u)$. The logistic choice probabilities allow us to interpret this weighted average as a conditional expectation, which we exploit to prove uniqueness and starshapedness.

**Uniqueness.** Define

$$C(\Delta u) := \frac{\int \rho'_\beta(\Delta u)\beta \Delta u\, dF(\beta)}{\int \rho'_\beta(\Delta u)\, dF(\beta)},$$

so (19) becomes $\Delta v = C(\Delta u)$. Since $\rho_\beta(x)$ is the CDF of Logistic$(0, \lambda/\beta)$, we have $\rho'_\beta(x) = g_{\lambda/\beta}(x)$, where $g_s$ denotes the density of Logistic$(0, s)$. Define independent random variables $\tilde{v} \sim$ Logistic$(0, \lambda)$ and $\tilde{\beta} \sim F$, and let $\tilde{u} := \tilde{v}/\tilde{\beta}$. Then

$$C(\Delta u) = \mathbb{E}[\tilde{v} \mid \tilde{u} = \Delta u].$$

Since $\log \tilde{u} = \log \tilde{v} - \log \tilde{\beta}$, $\log \tilde{u}$ is a location experiment of $\log \tilde{v}$ with noise $-\log \tilde{\beta}$. Since $\log \tilde{\beta}$ is strictly log-concave by assumption, the experiment satisfies the strict monotone likelihood ratio property. Hence the posterior over $\log \tilde{v}$ given $\log \tilde{u} = \log \Delta u$ is strictly increasing in first-order stochastic dominance in $\log \Delta u$, and so the posterior mean $C(\Delta u)$ is strictly increasing. Thus $\Delta v$ is uniquely determined by $\Delta u$. Since $Q$ maps $\Delta v \mapsto \Delta u$, $Q$ is unique, and by Lemma 5, so is $R$. Thus (19) characterizes the solution $Q$.[15]

**Property 2: Starshapedness.** By (19), $\Delta u = 0$ when $\Delta v = 0$, so $Q(0) = 0$. Moreover, since the term in parentheses in (19) is a weighted average of $\beta > 0$, we have $\Delta u > 0$ when $\Delta v > 0$, so $Q(\Delta v) > 0$ for $\Delta v > 0$. By Definition 2, it remains to show $Q(k\Delta v) > kQ(\Delta v)$ for $\Delta v > 0$

[15] Even without the assumption that $\log \beta$ is strictly log-concave, one obtains a weaker form of uniqueness. Under two actions, the objective (11) is strictly supermodular in $\Delta u$ and $\Delta v$, so the solutions $\Delta u$ are strictly increasing in the strong set order. Thus, $Q$ is non-decreasing and wherever non-unique, it has an upward jump. Since a non-decreasing function can have only countably many jump discontinuities, $Q$ and hence $R$ are unique almost everywhere on $\mathbb{R}$.

and $k > 1$. This is equivalent to $Q(\Delta v)/\Delta v$ being strictly increasing on $\mathbb{R}_{>0}$, or equivalently, $Q^{-1}(\Delta u)/\Delta u$ being strictly decreasing on $\mathbb{R}_{>0}$ (using that $Q$ is strictly increasing by the uniqueness proof). By (19), $Q^{-1} = C$, so

$$\frac{Q^{-1}(\Delta u)}{\Delta u} = \frac{C(\Delta u)}{\Delta u} = \mathbb{E}[\tilde{\beta} \mid \tilde{u} = \Delta u].$$

Since $\log \tilde{u} = \log \tilde{v} - \log \tilde{\beta}$, $\log \tilde{u}$ is a location experiment of $-\log \tilde{\beta}$ with noise $\log \tilde{v}$. The logistic distribution is strictly log-concave, which implies that $\log \tilde{v}$ has a strictly log-concave density. By the strict monotone likelihood ratio property, the posterior over $-\log \tilde{\beta}$ given $\log \tilde{u} = \log \Delta u$ is strictly increasing in first-order stochastic dominance in $\log \Delta u$. Since $\tilde{\beta}$ is decreasing in $-\log \tilde{\beta}$, the posterior over $\tilde{\beta}$ is strictly decreasing in first-order stochastic dominance, hence $\mathbb{E}[\tilde{\beta} \mid \tilde{u} = \Delta u]$ is strictly decreasing in $\Delta u$.

**Property 1: $R(x) \gtreqless 0 \Leftrightarrow x \gtreqless 0$.** Given $Q(x)$, the values $R(x)$ and $R(-x)$ minimize the socialization cost subject to $x + R(x) - R(-x) = Q(x)$:

$$\min_{R(x), R(-x) \in \mathbb{R}} \left( \int \rho_\beta(Q(x))\, dF(\beta) \right) \phi(R(x)) + \left( 1 - \int \rho_\beta(Q(x))\, dF(\beta) \right) \phi(R(-x)) \qquad (20)$$
$$\text{s.t.} \quad x + R(x) - R(-x) = Q(x).$$

We have $R(0) = 0$ from above. For $x > 0$, equation (19) implies $Q(x) - x > 0$ by $\beta < 1$. If $R(x) \leq 0$, then $R(-x) = R(x) - (Q(x) - x) < 0$. Since $\phi$ is strictly convex with minimum at 0 and $\phi'(0) = 0$, increasing both $R(x)$ and $R(-x)$ by a small $\varepsilon > 0$ preserves the constraint and decreases the objective. Hence $R(x) > 0$. The case $x < 0$ is analogous.

**Continuity.** By Lemma 3, we can restrict $\Delta u$ to lie in a compact set that depends continuously on $\Delta v$. Since the objective (10) is jointly continuous in $(\Delta u, \Delta v)$, Berge's maximum theorem and uniqueness imply $Q$ is continuous.

For $R$: using the constraint, rewrite (20) as minimizing over $R(x)$ alone:

$$\min_{R(x) \in \mathbb{R}} \left( \int \rho_\beta(Q(x))\, dF(\beta) \right) \phi(R(x)) + \left( 1 - \int \rho_\beta(Q(x))\, dF(\beta) \right) \phi(Q(x) - x - R(x)).$$

By Property 1, we can restrict attention to $R(x) \in [0, Q(x) - x]$ for $x \geq 0$ and $R(x) \in [Q(x) - x, 0]$ for $x \leq 0$. This compact-valued correspondence is continuous in $x$ since $Q$ is continuous. The objective is jointly continuous in $(x, R(x))$ since $Q$, $\rho_\beta$, and $\phi$ are continuous. By Berge's maximum theorem and uniqueness, $R$ is continuous. $\square$

## D Proposition 1

*Proof.* Define $\Psi(x, y) := Q(\bar{v}(x) - \bar{v}(y))$. Since $Q$ is skew-symmetric and continuous and $\bar{v}$ is continuous, $\Psi$ is skew-symmetric and continuous, so preferences have a generalized regret representation. Starshapedness and skew-symmetry together imply that $Q$ is strictly increasing. Since $\bar{v}$ and $Q$ are strictly increasing, $\Psi$ is strictly increasing in its first argument, establishing increasingness. For convexity, let $x > y > z$ and define $a := \bar{v}(x) - \bar{v}(y) > 0$ and $b := \bar{v}(y) - \bar{v}(z) > 0$. Then $\Psi(x, z) = Q(a + b)$ and $\Psi(x, y) + \Psi(y, z) = Q(a) + Q(b)$. For nonnegative functions on $[0, \infty)$ that vanish at 0, starshapedness implies superadditivity (Bruckner & Ostrow 1962), so $Q(a + b) > Q(a) + Q(b)$.[16] $\square$

## E Proposition 2

*Proof.* Loomes and Sugden (1982) show that the Allais paradox and the common-ratio effect are equivalent to strict superadditivity of $Q$,

$$\forall x_1, x_2 \colon 0 < x_2 < x_1 \to Q(x_1) - Q(x_1 - x_2) - Q(x_2) > 0, \tag{21}$$

and that simultaneous gambling and insurance is equivalent to

$$\forall x > 0 \colon Q\left(\tfrac{p}{1-p} x\right) \gtreqless \tfrac{p}{1-p} Q(x) \quad \text{if } p \gtreqless 0.5. \tag{22}$$

As established above, our definition of starshapedness implies strict superadditivity, hence (21). Condition (22) is equivalent to starshapedness: substituting $k = \frac{p}{1-p}$, we have $k > 1$ if and only if $p > 0.5$, and (22) becomes $Q(kx) > kQ(x)$ for $k > 1$. $\square$

## F Undistorted Choice under Equilibrium Beliefs

It is useful to recall the distinction between the attention stage (choice of a signal structure) and the decision-making stage (choice of an action conditional on the signal). A common theme in the principal-expert literature is that the principal might want to distort the expert's incentives at the decision-making stage to incentivize more information acquisition, as already pointed out by Demski and Sappington (1987). The following result shows that in our model, this is not necessary: under equilibrium beliefs, the agent's choice is undistorted.

**Observation 1.** *Under the beliefs that agents acquire in equilibrium, the agent's choice coincides with the principal's preferred action at that belief.*

*Proof.* Fix any type $\beta$. We show that the posterior $\gamma_\beta^{a_1}$ this type holds when choosing $a_1$ makes $a_1$ optimal for the principal. Since the principal's payoff from action $a$ under state $\omega$ is $v(a, \omega)$, we need to show

$$\mathbb{E}_{\gamma_\beta^{a_1}}[\Delta v] := \sum_{\omega \in \Omega} \gamma_\beta^{a_1}(\omega) \Delta v(\omega) > 0$$

where $\Delta v(\omega) = v(a_1, \omega) - v(a_2, \omega)$.

[16] Bruckner and Ostrow (1962) prove this for weak inequalities; the strict version follows by the same argument.

By symmetry, $p_\beta(a_1) = 1/2$, so the posterior is

$$\gamma_\beta^{a_1}(\omega) = \frac{\mu(\omega)\, p_\beta(a_1|\omega)}{p_\beta(a_1)} = 2\mu(\omega)\, p_\beta(a_1|\omega).$$

For $x \in \mathbb{R}$, let $\Omega_x = \{\omega \in \Omega : \Delta v(\omega) = x\}$ denote the set of states with payoff difference $x$. By symmetry, $\mu(\Omega_x) = \mu(\Omega_{-x})$ for all $x$. We pair such sets and show that each pair contributes positively to $\mathbb{E}_{\gamma^\beta_{a_1}}[\Delta v]$.

Let $x > 0$. Since $Q$ is strictly increasing with $Q(0) = 0$, we have $Q(x) > 0 > Q(-x)$, so for all $\omega \in \Omega_x, \omega' \in \Omega_{-x}$

$$p_\beta(a_1|\omega) = p_\beta(Q(x)) > \tfrac{1}{2} > p_\beta(-Q(x)) = p_\beta(a_1|\omega')$$

where, with slight abuse of notation, $p_\beta(x) = e^{\frac{\beta}{\lambda}x}/(1 + e^{\frac{\beta}{\lambda}x})$. The contribution of $\Omega_x$ and $\Omega_{-x}$ to $\mathbb{E}_{\gamma_\beta^{a_1}}[\Delta v]$ is

$$\sum_{\omega \in \Omega_x} \gamma_\beta^{a_1}(\omega)\Delta v(\omega) + \sum_{\omega' \in \Omega_{-x}} \gamma_\beta^{a_1}(\omega')\Delta v(\omega') = 2\mu(\Omega_x)\big(p_\beta(Q(x)) - p_\beta(-Q(x))\big) \cdot x > 0.$$

Summing over all $x > 0$, we obtain $\mathbb{E}_{\gamma_\beta^{a_1}}[\Delta v] > 0$. The argument for $a_2$ is analogous. $\square$

## G Proposition 3

We first establish a scaling property for the state-wise maximization problem, which applies to any number of actions.

**Lemma 8** (Scaling)**.** *Consider the state-wise maximization problem* (B) *with material payoffs $v(\cdot,\omega)$ and complexity $\lambda$. If $u(\cdot,\omega)$ is a solution, then $k \cdot u(\cdot,\omega)$ is a solution to the problem with payoffs $k \cdot v(\cdot,\omega)$ and complexity $k\lambda$, for any $k > 0$.*

*Proof.* The choice probabilities depend on $(u, \lambda)$ only through the ratio $\beta u/\lambda$:

$$p_\beta(a|u(\omega)) = \frac{e^{\frac{\beta}{\lambda}u(a,\omega)}}{\sum_{a' \in A} e^{\frac{\beta}{\lambda}u(a',\omega)}}.$$

Under complexity $k\lambda$, the choice probabilities at $ku$ satisfy $p_\beta(a|ku(\omega); k\lambda) = p_\beta(a|u(\omega); \lambda)$, so the choice probabilities are unchanged. The objective (B) under $(kv, k\lambda)$ evaluated at $ku$ is:

$$\int \left( k\lambda\, H(p_\beta(ku)) + \sum_{a \in A} p_\beta(a|ku) \cdot kv(a,\omega) \right) dF(\beta) = k \int \left( \lambda\, H(p_\beta(u)) + \sum_{a \in A} p_\beta(a|u) \cdot v(a,\omega) \right) dF(\beta).$$

Since scaling the objective by a positive constant preserves maximizers, $ku$ solves the problem under $(kv, k\lambda)$ if and only if $u$ solves it under $(v, \lambda)$. $\square$

*Proof of Proposition 3.* (i) By Lemma 8, scaling $(v, \lambda)$ to $(kv, k\lambda)$ scales the state-wise solution from $u$ to $ku$. The principal's objective and the agent's incentive constraint both scale by $k$, preserving maximizers. The symmetry constraint is also preserved since scaling all payoffs by the same constant maintains the distribution over payoff vectors. Therefore, if $u$ solves (P) under $(v, \lambda)$, then $ku$ solves (P) under $(kv, k\lambda)$.

Under two actions, $Q_\lambda$ maps $\Delta v$ to $\Delta u$, so $Q_{k\lambda}(k\Delta v) = k \cdot Q_\lambda(\Delta v)$. Setting $k = \lambda'/\lambda$ and $x = k\Delta v$ yields the scaling property.

(ii) The comparative static on $Q$ follows from (i) and starshapedness. For $\lambda' < \lambda$, let $k = \lambda'/\lambda < 1$. By the scaling property:

$$Q_{\lambda'}(x) = k\, Q_\lambda(x/k).$$

Since $1/k > 1$ and $x > 0$, starshapedness of $Q_\lambda$ implies $Q_\lambda(x/k)/(x/k) > Q_\lambda(x)/x$, so $Q_\lambda(x/k) > (1/k)\, Q_\lambda(x)$. Therefore:

$$Q_{\lambda'}(x) = k\, Q_\lambda(x/k) > k \cdot \frac{1}{k}\, Q_\lambda(x) = Q_\lambda(x).$$

For the comparative static on $|R(-x)|$, we use supermodularity. Recall from the proof of Theorem 1 that $R(x)$ and $R(-x)$ minimize the expected implementation cost subject to being consistent with $Q(x)$:

$$\begin{aligned} \min_{R(x), R(-x) \in \mathbb{R}} \quad & q \cdot \phi(R(x)) + (1-q) \cdot \phi(R(-x)) \\ \text{s.t.} \quad & R(x) - R(-x) = Q(x) - x, \end{aligned}$$

where $q := \int p_\beta(Q(x))\, dF(\beta)$ is the probability of taking the right action and $p_\beta(y) = e^{\frac{\beta}{\lambda} y}/(1+e^{\frac{\beta}{\lambda} y})$. Substituting the constraint $R(x) = Q(x) - x + R(-x)$, the problem reduces to minimizing over $R(-x)$:

$$\Phi(R(-x); q, Q(x)) := q \cdot \phi(Q(x) - x + R(-x)) + (1-q) \cdot \phi(R(-x)).$$

When $\lambda$ decreases to $\lambda'$, both $Q(x)$ increases (as just shown) and $q$ increases, since $p_\beta$ is strictly increasing in its argument and $q$ depends on $Q(x)/\lambda$. We show that $\Phi$ has strictly increasing differences in $(R(-x), q)$ and in $(R(-x), Q(x))$. Recall from Theorem 1 that at the optimum, $R(-x) < 0$ and $R(x) = Q(x) - x + R(-x) > 0$.

For the cross-partial in $(R(-x), q)$:

$$\frac{\partial^2 \Phi}{\partial R(-x)\, \partial q} = \phi'(Q(x) - x + R(-x)) - \phi'(R(-x)).$$

Since $\phi$ is strictly convex and minimized at 0, we have $\phi'(y) > 0$ for $y > 0$ and $\phi'(y) < 0$ for $y < 0$. Thus $\phi'(Q(x) - x + R(-x)) > 0$ and $\phi'(R(-x)) < 0$, so the cross-partial is strictly positive.

For the cross-partial in $(R(-x), Q(x))$:

$$\frac{\partial^2 \Phi}{\partial R(-x)\, \partial Q(x)} = q \cdot \phi''(Q(x) - x + R(-x)) > 0,$$

since $q > 0$ and $\phi'' > 0$ by strict convexity.

By Topkis' theorem, the optimal $R(-x)$ is strictly decreasing in both $q$ and $Q(x)$. Since $R(-x) < 0$, this means $|R(-x)|$ is strictly increasing in both parameters, and hence strictly increasing when $\lambda$ decreases. □

## H Proposition 4

*Proof.* We cannot apply upper hemicontinuity from Berge's maximum theorem because our objective is not defined at $\lambda = 0$ or $\lambda = \infty$. Instead, we work directly with the first-order conditions for optimality.

Since the principal's problem separates by state, it suffices to prove convergence for the state-wise problem. Fix $\omega \in \Omega$ and adopt the shorthand $u_i := u(a_i, \omega)$ and $v_i := v(a_i, \omega)$ for $i = 1, \ldots, n$. By Assumption 4, assume $v_1 < \cdots < v_n$. By Lemma 2, $u_1 < \cdots < u_n$. Normalize $u_n = v_n = 0$ and define $u := (u_1, ..., u_{n-1})$ and $v := (v_1, ..., v_{n-1})$. Since we vary $\lambda$, we make the dependence on $\lambda$ explicit and write $p^i_{\beta,\lambda}(u)$ for the choice probability $p_\beta(a_i|\omega)$.

The first-order condition for the state-wise maximization problem with respect to $u_i$, derived from equations (12)–(14), is

$$\int \left( \frac{\beta}{\lambda} p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))(v_i - \beta u_i) - \sum_{j=1, j\neq i}^{n-1} p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u)(v_j - \beta u_j) \right) dF(\beta) = 0. \tag{23}$$

We prove the two limits separately, starting with the simpler $\lambda \to \infty$ case.

**Case $\lambda \to \infty$.** *Step 1: Limiting system of first-order conditions.* We multiply the first-order conditions (23) by $\lambda$ and write them in matrix form as $\mathcal{A}(u, \lambda)v = \mathcal{B}(u, \lambda)u$, where $\mathcal{A}$ and $\mathcal{B}$ are $(n-1) \times (n-1)$ matrices with entries

$$\mathcal{A}_{ij}(u, \lambda) = \begin{cases} \int \beta \, p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u)) \, dF(\beta) & \text{if } i = j \\ -\int \beta \, p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u) \, dF(\beta) & \text{if } i \neq j \end{cases}$$

$$\mathcal{B}_{ij}(u, \lambda) = \begin{cases} \int \beta^2 \, p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u)) \, dF(\beta) & \text{if } i = j \\ -\int \beta^2 \, p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u) \, dF(\beta) & \text{if } i \neq j \end{cases}$$

As $\lambda \to \infty$, we have $p^i_{\beta,\lambda}(u) \to 1/n$ for all $i$, $\beta$, and $u$. Thus

$$p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u)) \to \frac{1}{n} \cdot \frac{n-1}{n} = \frac{n-1}{n^2}, \qquad p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u) \to \frac{1}{n^2}.$$

The matrices converge to

$$\mathcal{A}(u,\lambda) \to \mathbb{E}[\beta] \cdot M,$$
$$\mathcal{B}(u,\lambda) \to \mathbb{E}[\beta^2] \cdot M,$$

where $M$ is the $(n-1)\times(n-1)$ matrix with $(n-1)/n^2$ on the diagonal and $-1/n^2$ off the diagonal. The limiting system of first-order conditions is

$$\mathbb{E}[\beta] \cdot Mv = \mathbb{E}[\beta^2] \cdot Mu.$$

Since $M$ is invertible (it is strictly diagonally dominant), this simplifies to $v_i = (\mathbb{E}[\beta^2]/\mathbb{E}[\beta])u_i$ for each $i = 1, \ldots, n-1$. The unique solution is $u_i^* := (\mathbb{E}[\beta]/\mathbb{E}[\beta^2])v_i$.

*Step 2: Convergence.* Reparametrize with $l = 1/\lambda$, so $\lambda \to \infty$ corresponds to $l \to 0$. With slight abuse of notation, write the matrices as functions of $l$ and the system of first-order conditions as $F(u,v,l) := \mathcal{A}(u,l)v - \mathcal{B}(u,l)u = 0$, which is jointly continuous in $(u,v,l)$ including at $l = 0$. At $l = 0$, the limiting system has the unique solution $u^*(v) := (\mathbb{E}[\beta]/\mathbb{E}[\beta^2])v$.

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We show that solutions $u$ to the $F(u,v,l) = 0$ converge to $u^*(v)$ as $l \to 0$, uniformly over $v$ in any compact set $V \subseteq \mathbb{R}^{n-1}$. By the proof of Lemma 3, solutions lie in a compact set $K_V$ depending only on $V$. Joint continuity of $F$ implies that the solution correspondence has closed graph. Together with compactness of $K_V$, this implies upper hemicontinuity. Since the limiting system has a unique solution $u^*(v)$, which is continuous in $v$, upper hemicontinuity implies that solutions converge to $u^*(v)$, uniformly over $v \in V$.

**Case $\lambda \to 0$.** *Step 1: Normalizing the first-order conditions.* Since $u_n = 0 > u_j$ for all $j < n$, $p^n_{\beta,\lambda}(u) \to 1$ and $p^j_{\beta,\lambda}(u) \to 0$ for all $j < n$ as $\lambda \to 0$. Dividing the first-order condition (23) for row $i$ by $\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1-p^i_{\beta,\lambda}(u))\,dF(\beta)$, we obtain:

$$\begin{aligned} v_i &+ \sum_{j\neq i,n} \frac{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)p^j_{\beta,\lambda}(u)\,dF(\beta)}{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1-p^i_{\beta,\lambda}(u))\,dF(\beta)} v_j \\ &= \frac{\int(\beta^2/\lambda)p^i_{\beta,\lambda}(u)(1-p^i_{\beta,\lambda}(u))\,dF(\beta)}{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1-p^i_{\beta,\lambda}(u))\,dF(\beta)} u_i + \sum_{j\neq i,n} \frac{\int(\beta^2/\lambda)p^i_{\beta,\lambda}(u)p^j_{\beta,\lambda}(u)\,dF(\beta)}{\int(\beta/\lambda)p^i_{\beta,\lambda}(u)(1-p^i_{\beta,\lambda}(u))\,dF(\beta)} u_j. \end{aligned}$$

We show: (i) the coefficients on $v_j$ and $u_j$ for $j \neq i,n$ vanish as $\lambda \to 0$, and (ii) the coefficient on $u_i$ converges to $\underline{\beta}$.

*Step 2: Off-diagonal coefficients vanish.* Consider the coefficient on $v_j$ for $j \neq i, n$:

$$\frac{\int (\beta/\lambda) p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u)\, dF(\beta)}{\int (\beta/\lambda) p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)} = \frac{\int \frac{p^j_{\beta,\lambda}(u)}{1 - p^i_{\beta,\lambda}(u)} \cdot (\beta/\lambda) p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)}{\int (\beta/\lambda) p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)}.$$

This is a weighted average of $p^j_{\beta,\lambda}(u)/(1 - p^i_{\beta,\lambda}(u))$. Since $u_j < 0$:

$$p^j_{\beta,\lambda}(u) = \frac{e^{\beta u_j/\lambda}}{\sum_{m=1}^n e^{\beta u_m/\lambda}} \leq e^{\beta u_j/\lambda} \leq e^{\underline{\beta} u_j/\lambda} \to 0$$

as $\lambda \to 0$, uniformly over $\beta \in \text{supp}(\beta)$. For the denominator:

$$1 - p^i_{\beta,\lambda}(u) \geq p^n_{\beta,\lambda}(u) = \frac{1}{\sum_{m=1}^n e^{\beta u_m/\lambda}} \geq \frac{1}{n},$$

since each term in the sum is at most 1 (as $u_m \leq u_n = 0$ for all $m$). Thus $p^j_{\beta,\lambda}(u)/(1 - p^i_{\beta,\lambda}(u)) \leq n \cdot e^{\underline{\beta} u_j/\lambda} \to 0$ uniformly over $\beta$. Since the coefficient is a weighted average of a quantity converging uniformly to 0, the coefficient vanishes. The same argument applies to the coefficient on $u_j$.

*Step 3: Diagonal coefficient converges to $\underline{\beta}$.* The coefficient on $u_i$ is:

$$\frac{\int (\beta^2/\lambda) p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)}{\int (\beta/\lambda) p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))\, dF(\beta)} = \frac{\int g_\lambda(\beta)\beta\, dF(\beta)}{\int g_\lambda(\beta)\, dF(\beta)},$$

where $g_\lambda(\beta) := (\beta/\lambda) p^i_{\beta,\lambda}(u)(1 - p^i_{\beta,\lambda}(u))$. Fix $\varepsilon > 0$ and decompose:

$$\frac{\int g_\lambda(\beta)\beta\, dF(\beta)}{\int g_\lambda(\beta)\, dF(\beta)} = \frac{A(\lambda) + B(\lambda)}{1 + C(\lambda)},$$

where

$$A(\lambda) := \frac{\int_{\underline{\beta}}^{\underline{\beta}+\varepsilon} g_\lambda(\beta)\beta\, dF(\beta)}{\int_{\underline{\beta}}^{\underline{\beta}+\varepsilon} g_\lambda(\beta)\, dF(\beta)}, \quad B(\lambda) := \frac{\int_{\underline{\beta}+\varepsilon}^{1} g_\lambda(\beta)\beta\, dF(\beta)}{\int_{\underline{\beta}}^{\underline{\beta}+\varepsilon} g_\lambda(\beta)\, dF(\beta)},$$
$$C(\lambda) := \frac{\int_{\underline{\beta}+\varepsilon}^{1} g_\lambda(\beta)\, dF(\beta)}{\int_{\underline{\beta}}^{\underline{\beta}+\varepsilon} g_\lambda(\beta)\, dF(\beta)}.$$

We have $\underline{\beta} \leq A(\lambda) \leq \underline{\beta} + \varepsilon$ because $A(\lambda)$ is a weighted average of $\beta$ over $[\underline{\beta}, \underline{\beta} + \varepsilon]$. To show $B(\lambda) \to 0$ and $C(\lambda) \to 0$, we establish an exponential bound on $g_\lambda(\beta)$. For $\beta > \underline{\beta} + \varepsilon$:

$$\frac{p^i_{\beta,\lambda}(u)}{p^i_{\underline{\beta}+\varepsilon,\lambda}(u)} = e^{(\beta - (\underline{\beta}+\varepsilon))u_i/\lambda} \cdot \frac{\sum_{m=1}^n e^{(\underline{\beta}+\varepsilon)u_m/\lambda}}{\sum_{m=1}^n e^{\beta u_m/\lambda}} \leq n \cdot e^{(\beta - (\underline{\beta}+\varepsilon))u_i/\lambda},$$

where the first factor decays exponentially in $\beta$ since $u_i < 0$, and the second factor is bounded by

$n$. Combined with $(1-p^i_{\beta,\lambda}(u))/(1-p^i_{\underline{\beta}+\varepsilon,\lambda}(u)) \le n$ and $\beta/(\underline{\beta}+\varepsilon) \le 1/\underline{\beta}$:

$$g_\lambda(\beta) \le \frac{n^2}{\underline{\beta}} \cdot g_\lambda(\underline{\beta}+\varepsilon) \cdot e^{(\beta-\underline{\beta}-\varepsilon)u_i/\lambda}.$$

Thus:

$$B(\lambda) \le \frac{n^2/\underline{\beta}}{F(\underline{\beta}+\varepsilon)} \int_{\underline{\beta}+\varepsilon}^1 e^{(\beta-\underline{\beta}-\varepsilon)u_i/\lambda} \beta \, dF(\beta).$$

The integrand converges pointwise to 0 for all $\beta > \underline{\beta}+\varepsilon$ and $u < 0$, and is bounded by 1. By dominated convergence, $B(\lambda) \to 0$. Similarly, $C(\lambda) \to 0$.

Thus $\limsup_{\lambda\to 0} \frac{\int g_\lambda(\beta)\beta \, dF(\beta)}{\int g_\lambda(\beta)\, dF(\beta)} \le \underline{\beta}+\varepsilon$. Since $\varepsilon > 0$ was arbitrary and the ratio is bounded below by $\underline{\beta}$, the limit equals $\underline{\beta}$.

*Step 4: Solutions remain bounded away from* 0. The convergence arguments in Steps 2–3 require $u_i$ to be bounded away from 0 for the exponential bounds to apply uniformly. We now establish this.

Suppose, for contradiction, that along some sequence $\lambda_k \to 0$, we have solutions to the first-order conditions $u(\lambda_k) \to \bar{u}$ with $\bar{u}_i = 0$ for some $i < n$. Let $I := \{i < n : \bar{u}_i = 0\}$. By Lemma 2, $u_j \le 0$ for all $j < n$ and all $\lambda$, so $\bar{u}_j \le 0$ for all $j < n$.

Consider $u_i(\lambda_k)/\lambda_k$ for $i \in I$. Either:

*Subcase (a): $u_i(\lambda_k)/\lambda_k$ is unbounded below for some $i \in I$.* Then $u_i(\lambda_{k_j})/\lambda_{k_j} \to -\infty$ along some subsequence. The bounds from Step 2 give $p^i_{\beta,\lambda}(u) \le e^{\beta u_i(\lambda_{k_j})/\lambda_{k_j}} \to 0$. The arguments from Steps 2–3 apply, and the normalized first-order condition converges to $v_i = \underline{\beta} \cdot \bar{u}_i$. This gives $\bar{u}_i = v_i/\underline{\beta} \ne 0$ contradicting $\bar{u}_i = 0$.

*Subcase (b): $u_i(\lambda_k)/\lambda_k$ is bounded below for all $i \in I$.* Then $u_i(\lambda_k)/\lambda_k \in [-M, 0]$ for some $M > 0$. Extract a subsequence along which $u_i(\lambda_{k_j})/\lambda_{k_j} \to c_i \in [-M, 0]$ for all $i \in I$. For $i \notin I$, we have $\bar{u}_i < 0$, so $u_i/\lambda_k \to -\infty$, hence $p^i_{\beta,\lambda}(u) \to 0$.

Rewrite the first-order condition (23) for $i \in I$, multiplied by $\lambda$:

$$\sum_{j \ne i} \int \beta \, p^i_{\beta,\lambda}(u) p^j_{\beta,\lambda}(u) \Big[(v_i - \beta u_i) - (v_j - \beta u_j)\Big] dF(\beta) = 0. \tag{24}$$

The integrand is uniformly bounded: $\beta \le 1$, probabilities lie in $[0,1]$, and $(v_i - \beta u_i) - (v_j - \beta u_j)$ is bounded since $u_i, u_j$ lie in a compact set. By dominated convergence, we can pass to the limit under the integral. As $\lambda_{k_j} \to 0$: terms with $j \notin I \cup \{n\}$ vanish since $p^j_{\beta,\lambda}(u) \to 0$. For $j \in I \cup \{n\}$, we have $u_j \to 0$, so $(v_i - \beta u_i) - (v_j - \beta u_j) \to v_i - v_j$. The limiting choice probabilities $\bar{p}^i, \bar{p}^j > 0$ for $i, j \in I \cup \{n\}$ are well-defined since $u_m/\lambda \to c_m$ is bounded for $m \in I \cup \{n\}$. Since the left-hand side of (24) equals zero for each $\lambda_{k_j}$, the limit must also equal zero:

$$\sum_{j \in I \cup \{n\}, j \ne i} \int \beta \, \bar{p}^i \, \bar{p}^j \, (v_i - v_j) \, dF(\beta) = 0.$$

Let $i^* := \min I$ be the smallest index in $I$. Since $v_1 < \cdots < v_{n-1}$, we have $v_{i^*} < v_j$ for all $j \in I \cup \{n\}$ with $j \neq i^*$. Thus every term in the sum for $i = i^*$ satisfies $v_{i^*} - v_j < 0$, making the left-hand side strictly negative, which is a contradiction.

*Step 5: Convergence.* We show that solutions converge to $u^*(v) := v/\underline{\beta}$ as $\lambda \to 0$, uniformly over $v$ in any compact set $V \subseteq \mathbb{R}^{n-1}$ with $v < 0$. Fix such a $V$. By Lemma 3, solutions lie in a compact set $K_V$ depending only on $V$.

Consider any sequence $(\lambda^k, v^k)$ with $\lambda^k \to 0$ and $v^k \to v^* \in V$, and let $u^k$ be a solution at $(\lambda^k, v^k)$. Since $u^k \in K_V$, there exists a convergent subsequence $u^{k_j} \to \bar{u}$. By Step 4 applied to $v^*$ (the argument applies verbatim to sequences with $v^k \to v^* \in V$), we have $\bar{u}_i < 0$ for all $i < n$. Since $u^{k_j} \to \bar{u}$ with $\bar{u}_i < 0$, the solutions $u^{k_j}$ are bounded away from 0 for large $j$. The coefficient bounds from Steps 2–3 then apply along this subsequence: the off-diagonal coefficients vanish and the diagonal coefficient converges to $\underline{\beta}$. Taking limits in the normalized first-order conditions yields $v_i^* = \underline{\beta} \cdot \bar{u}_i$, giving $\bar{u} = v^*/\underline{\beta} = u^*(v^*)$. Since every convergent subsequence has the same limit and solutions $u^k$ lies in a compact set, $u^k \to u^*(v^*)$.

For uniform convergence, suppose not: then there exist $\varepsilon > 0$, $\lambda^k \to 0$, $v^k \in V$, and solutions $u^k$ at $(\lambda^k, v^k)$ with $|u^k - u^*(v^k)| \geq \varepsilon$. Since $V$ is compact, extract a subsequence with $v^{k_j} \to v^* \in V$. By the above, $u^{k_j} \to u^*(v^*)$. By continuity of $u^*$, $u^*(v^{k_j}) \to u^*(v^*)$. Thus $|u^{k_j} - u^*(v^{k_j})| \to 0$, contradicting $|u^{k_j} - u^*(v^{k_j})| \geq \varepsilon$. $\square$

# VI Appendix B: Incomplete Feedback

## A Incomplete Feedback

So far, we have assumed that the principal can choose emotions as an arbitrary function of the action and realized state. In many decision contexts, however, ex post only the payoff is revealed but not the full state. For example, if the agent pursues some economic project, the realized return might be observable but not the return had he pursued another project. In particular, when only the realized payoff $v(a,\omega)$ is learned but not the state $\omega$, the principal cannot in general infer the payoffs of counterfactual actions. Thus, she cannot implement the general regret solution from the previous section. This section generalizes the principal's problem to account for incomplete feedback under the simplifying assumption of a known bias.

As noted earlier, under known bias it is possible to implement the first-best attention strategy by offering a contract that scales up the payoff-differences by $1/\beta$. However, to find the emotion scheme $m$ with minimal variance, we need to determine *all* emotion schemes that implement the optimal attention strategy. Below, we argue how this exercise teaches us something about the nature of the optimal mechanism when the first-best attention strategy cannot be implemented due to an uncertain bias.

Ex-post feedback also plays an important role in the study of regret and disappointment. While classical regret theory (Bell 1982; Loomes and Sugden 1982) does not take into account feedback, several experiments have documented that regret affects decisions more if feedback is expected (Zeelenberg 1999). To accommodate for that evidence, Humphrey (2004) introduces a theory of feedback-conditional regret which employs two regret-rejoicing functions, one for complete feedback and one for incomplete feedback. Gabillon (2020) proposes a regret theory for arbitrary feedback structures, where the regret-rejoicing function is applied to the difference between the actual payoff and the highest counterfactual payoff, given the posterior beliefs after feedback. Evidence from Camille et al. (2004) based on self-reported emotions, skin conductance responses, and choices suggests, however, that under partial feedback (only the payoff of the chosen action is learned) primarily disappointment and elation are experienced, while under complete feedback mainly regret and rejoicing are experienced. We are not aware of theories that combine regret and disappointment depending on the feedback structure.

To incorporate feedback into our model, we assume that the principal can choose emotions only as a function of the feedback that is received ex post. Formally, we assume the emotion scheme $m$ is measurable with respect to a feedback structure $\mathcal{F}\colon A\times\Omega\to Z$ where $Z$ is a message space, that is,

$$\forall a,a'\in A,\omega,\omega'\in\Omega\colon \mathcal{F}(a,\omega)=\mathcal{F}(a',\omega')\to m(a,\omega)=m(a',\omega').$$

We consider partitional feedback structures that reveal the payoff $v(a,\omega)$, the action $a$, and a partition $\mathcal{P}$ of the state space.[17] In one extreme case, the feedback structure reveals the state fully (the finest partition; complete feedback) and in the other extreme, only the payoff is revealed (the coarsest partition; partial feedback). In intermediate cases, some information about the state is revealed, such as general market conditions, while more specific information, such as how each particular economic project would have done, is not revealed.

In some decision-problems, however, the agent can back out more information about the state from the material payoff and the action. To rule out such cases, we make an additional assumption that the message space is a "connected" on each partition. To formulate this condition, we endow the message space, $Z$, with a symmetric relation $\sim$, which makes $Z$ an undirected graph where $\sim$ stands for the edges of the graph. We define $z \sim z'$ if and only if there exists $\omega \in \Omega$ and $a, a' \in A$ such that $z = \mathcal{F}(a, \omega)$ and $z' = \mathcal{F}(a', \omega)$. Thus, two messages are directly connected by an edge if they are observed under the same state. Because the state-wise payoff-differences determine the incentives for attention, this is the relevant notion of connectedness that ensures that the emotions of the two messages have to be determined jointly. More specifically, the following condition ensure that the set of messages consistent with a partition cannot be divided into disjoint subsets such that emotions can be chosen independently on each subset.

**Assumption 5** (Partitional Feedback Structure)**.** *There is a partition $\mathcal{P}$ of $\Omega$ such that*

$$\mathcal{F}(a, \omega) = (v(a, \omega), a, \mathcal{P}(\omega))$$

*where $\mathcal{P}(\omega)$ is the partition cell that contains $\omega$. We define $Z$ as the image of $\mathcal{F}$ in $\mathbb{R} \times A \times \mathcal{P}$.*

*For each partition cell $P \in \mathcal{P}$, the subset of the message space that is consistent with this partition cell, $\{(v, a, P) \in Z | v \in \mathbb{R}, a \in A\}$, is a connected subgraph of $Z$.*

For simplicity, we analyze the model under a known bias $\beta \in (0, 1)$. Thus, we consider a pure hidden-action problem without hidden information.

**Assumption 6.** *The distribution of $\beta \in (0, 1)$ is degenerate.*

Under known $\beta$, we can generalize the model in two dimensions. First, we drop the symmetry assumption on $v$ and the restriction to mechanisms with symmetric $v + m$. Second, we generalize the entropy-based attention cost to any posterior-separable cost function. To be coherent with the notation above, we continue to define the attention cost in terms of the state-dependent stochastic choice function.[18] For each state-dependent stochastic choice function $p \colon \Omega \to \Delta(A)$, let $\tau(p) \in \Delta(\Delta(\Omega))$ denote the distribution over posteriors $\gamma \in \Delta(\Omega)$ induced by $p$ via Bayesian updating.

[17] A partition of the state space is a set of mutually exclusive and exhaustive subsets $\mathcal{P} = \{P_1, ..., P_n\}$ of $\Omega$. In the Appendix, we show that the resulting emotions are essentially of the same form, when we allow for other deterministic feedback structures that reveal at least the payoff. Further, none of the results below would change if the action was not revealed.

[18] This is without loss if the function $H$ below is strictly concave. As is well-known, if we defined the attention cost as a function of the acquired signal structure, we can without loss identify the signal with the action under strictly concave $H$.

**Definition 10** (Posterior-Separable attention cost)**.** *The attention cost $c$ is posterior-separable if*

$$c(p) = \mathbb{E}_{\tau(p)}[H(\mu) - H(\gamma)]$$

*where $H \colon \Delta(\Omega) \to \mathbb{R}$ is strictly concave and differentiable on $\Delta(\Omega)$.*

Posterior-separable attention costs arise naturally from several foundations, including information theory (Sims 2003), sequential sampling (Morris and Strack 2019; Hébert and Woodford 2023), and constant marginal cost of experimentation (Pomatto, Strack, and Tamuz 2023). The function $H$ can be understood as a measure of uncertainty of a distribution, with entropy as an example. Thus, under a posterior-separable cost, the agent incurs a cost proportional to the expected reduction of the uncertainty of their belief. That $H$ is strictly concave ensures that more information (in the Blackwell sense) is more costly.

**Assumption 7.** *The attention cost $c(p)$ is posterior separable.*

Given incomplete feedback and the absence of hidden information, the new principal's problem $\mathrm{P}^{\mathrm{inc}}$ is to pick the solution with minimal variance of emotions to:

$$\max_{m,p} -c(p) + \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega)p(a|\omega)v(a,\omega) \tag{$\mathrm{P}^{\mathrm{inc}}$}$$

$$\text{s.t.}$$

$$p \in \arg\max_{\tilde{p}} -c(\tilde{p}) + \beta \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega)\tilde{p}(a|\omega)\big(v(a,\omega) + m(a,\omega)\big) \tag{IC}$$

$$\sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega)p(a|\omega)m(a,\omega) = 0 \tag{BC}$$

$$\forall a, a' \in A, \omega, \omega' \in \Omega \colon \mathcal{F}(a,\omega) = \mathcal{F}(a',\omega') \to m(a,\omega) = m(a',\omega') \tag{F}$$

We introduce the following class of solutions, which we interpret below as a combination of regret and disappointment. To simplify the exposition, we assume that the first-best attention strategy $p$, which maximizes the principal's objective, is unique. Further, we define the consideration set $S \subseteq A$ as the set of actions that have positive probability according to the first-best attention strategy $p$, formally $S = \{a \in A | \sum_{\omega\in\Omega} \mu(\omega)p(a|\omega) > 0\}$.

**Definition 11** (Linear Regret-Disappointment Solution)**.** *A solution $(m,p)$ to the principal's problem $\mathrm{P}^{\mathrm{inc}}$ is a linear regret-disappointment solution if for all $a \in S, \omega \in \Omega$:*

$$m(a,\omega) = \left(\frac{1}{\beta} - 1\right)\big(v(a,\omega) - v(\mathcal{P}(\omega))\big) \tag{25}$$

*where $v(\mathcal{P}(\omega))$ is the expected payoff conditional on partition element $\mathcal{P}(\omega)$ with respect to the first-best attention strategy $p$,*

$$v(\mathcal{P}(\omega)) = \frac{1}{\mu(\mathcal{P}(\omega))} \sum_{\omega'\in\mathcal{P}(\omega)}\sum_{a\in S} \mu(\omega')p(a|\omega')v(a,\omega').$$

Such a solution has the natural interpretation as setting for each partition cell a reference point or benchmark $v(\mathcal{P}(\omega))$, which corresponds to the expected payoff the agent should achieve – according to the first-best attention strategy – conditional on that partition cell.[19] Note that in contrast to other models of reference points (Kahneman and Tversky 1979; Gul 1991; Kőszegi and Rabin 2006), the optimal gain-loss utility here is linear.

It is instructive to compare (25) to the regret theory by Gabillon (2020), where regret depends on the difference between the realized payoff and the highest payoff that could have been achieved through some action given the ex-post feedback about the state. As in their proposal, the reference point is updated on feedback about the state.[20] However, in our model, the reference point is not the highest attainable counterfactual payoff but instead the *expected* payoff conditional on ex-post feedback. Because this expected payoff is also affected by the payoffs of the chosen action *had different states realized*, one can interpret the emotional payoff as a combination of regret and disappointment. Next, we make this intuition more clear.

Another, mathematically equivalent, formulation of (25) relates the emotional payoff to regret and disappointment. Let $\gamma\colon A \to \Delta(\Omega)$ map each action to the posterior associated with it via Bayesian updating, which we write $\gamma(\omega|a)$. We can define, *conditional on partition cell* $\mathcal{P}(\omega)$, the probability, $p(a|\mathcal{P}(\omega))$, and the expected payoff of, $v(a|\mathcal{P}(\omega))$, of $a$,

$$\begin{aligned} p(a|\mathcal{P}(\omega)) &:= \sum_{\omega' \in \mathcal{P}(\omega)} \frac{\mu(\omega')}{\mu(\mathcal{P}(\omega))} p(a|\omega') \\ v(a|\mathcal{P}(\omega)) &:= \sum_{\omega' \in \mathcal{P}(\omega)} \frac{\gamma(\omega'|a)}{\gamma(\mathcal{P}(\omega)|a)} v(a,\omega'). \end{aligned}$$

Given this notation, we can rearrange terms and rewrite equation (25) as

$$\begin{aligned} m(a,\omega) = \sum_{a \neq a'} p(a'|\mathcal{P}(\omega)) \left(\frac{1}{\beta} - 1\right) \big(v(a,\omega) - v(a'|\mathcal{P}(\omega))\big) \\ + p(a|\mathcal{P}(\omega)) \left(\frac{1}{\beta} - 1\right) \big(v(a,\omega) - v(a|\mathcal{P}(\omega))\big). \end{aligned} \tag{26}$$

The term in the first line can be interpreted as regret and rejoicing. The emotion depends on the weighted difference between the actual payoff and the expected payoffs of all other actions $a'$, conditional on partition cell $\mathcal{P}(\omega)$. The expected payoff of each action is computed with respect to the first-best attention strategy. The regret-like comparisons to different actions are weighted by the probabilities of the respective actions, again conditional on the partition element $\mathcal{P}(\omega)$.

This is in contrast to a widely used theory of regret where the comparison is with respect to the ex-post optimal action (Quiggin 1994; Sarver 2008), which allows for regret only and no

[19] In the Appendix, we show that for non-quadratic costs of emotions, an analogous solution with a different reference point $v$ is optimal.

[20] To be precise, in (25), the reference point is updated only on feedback about the state that is action-independent, namely what is learned through the partition of the state space. It is not updated on what is learned from the realized payoff $v(a,\omega)$.

rejoicing. This proposal is arguably extreme when the agent has a large choice set and does not seem psychologically plausible when the ex-post optimal action was not even considered. Moreover, in some relevant applications as in finance, under no short-sale constraints the ex-post optimal return is infinite (Qin 2020). In contrast, our prediction corresponds to the proposal of Loomes and Sugden (1982) to generalize regret theory to multiple actions based on such a weighted sum of comparisons to *all* other actions, not just the ex-post optimal one. They highlight the importance of providing a theory of the appropriate action weights. Sugden (1986) writes that

> *"Any plausible theory of such weights must, I think, take some account of the extent to which an action is a serious candidate for choice or, to put it slightly differently, of the extent to which the individual could sensibly blame himself for not having chosen it."*

In our contexts, the action weights are naturally provided by the state-dependent choice probabilities under the first-best attention strategy. These choice probabilities capture with what probability the agent should have taken some action under the realized state. This implies that regret and rejoicing do not depend on actions outside of the consideration set, which are taken with probability 0. This captures that as Sugden (1986) writes, only actions that are "serious candidates" or "real options" should be sources of regret and rejoicing. While regret and rejoicing are linear here and thus predict no violations of expected utility axioms from choice, linear regret still has effects on menu choice (Sarver 2008).

The term in the second line of (26) can be interpreted as disappointment and elation. The emotion depends on the different between the payoff $v(a, \omega)$ and the expected payoff of the chosen action. The expected payoff of the chosen action is computed with respect to the posterior that the agent holds when taking the action. However, in contrast to standard models of disappointment and elation (Bell 1985; Loomes and Sugden 1986; Gul 1991), the expected payoff is conditioned on what is learned about the state. The weight on disappointment and elation is the probability of taking the actual action conditional on the partition cell.

**Theorem 3.** *Under Assumptions 5 to 6, any solution of* $\mathrm{P}^{\mathrm{inc}}$ *is a linear regret-disappointment solution and induces the first-best attention strategy.*

Because we assumed that the bias is known, the optimal mechanism can perfectly offset it by scaling up the utility differences by $1/\beta$. Using a result in Whitmeyer and Zhang (2022), which builds on the Lagrangian Lemma of Caplin, Dean, and Leahy (2022), we show that such inflating of the utility differences is also *necessary*, in order to implement the first-best attention strategy. The remaining goal of the principal is to reduce the variance of emotions, while scaling up the utility differences. This goal is constrained by the feedback structure. Under complete feedback, the second line of (26) is 0, and only regret is used. However, for incomplete feedback structures, both lines of (26) are generally non-zero and accordingly the optimal emotions can be interpreted as a combination of regret and disappointment.

Disappointment and elation are used because, under incomplete feedback, the choice of all emotional utilities consistent with some partition element of states is interrelated (Appendix, Lemma

9). Thus, all payoffs under that partition element, including the payoffs of the chosen action *had another state realized*, affect the optimal emotions under some feedback signal.[21]

Thus, in our model, disappointment and elation can be thought of as *second-best* motivators for attention when complete feedback is not available. They are second-best because they lead to a higher variance of emotions than using regret and rejoicing.[22]

One might be surprised that disappointment and elation motivate more attention because they depend on the realization of the state, which the agent cannot affect. By contrast, regret and rejoicing seem to depend more directly on the agent's action, which perhaps more intuitively incentivizes attention. However, although the agent cannot choose the unconditional distribution of states, he can implicitly choose the distribution of states *conditional* on an action by paying attention. Indeed, this is the perspective of attention taken by the posterior approach, in which the agent acquires posteriors over states conditional on each taken action (Caplin and Dean 2013). This goes to say that, like regret and rejoicing, the emotions of disappointment and elation penalize bad outcomes and reward good outcomes and thus raise the return to attention. As argued above, regret and rejoicing raise the return to attention in the optimal way but under constraints through the feedback structure, disappointment and elation have a role to play.

## B Proof of Theorem 3

Because the feedback structure reveals the material payoff by Assumption 5, the principal can choose the utility $u(z)$ for each message $z = (v, a, P) \in Z$. For $z = (v, a, P)$, we write $v(z)$ and $a(z)$ for the first and third component, respectively. The following Lemma uses of the fact that the optimal attention strategy under posterior-separable attention costs depends on the state-wise utility differences as shown in Whitmeyer and Zhang (2022).

**Lemma 9.** *A first-best attention strategy is induced only if for all elements $z_1, z_2$ of $Z$ that are directly connected by an edge and have $a(z_1), a(z_2) \in S$,*

$$u(z_1) - u(z_2) = \frac{1}{\beta}(v(z_1) - v(z_2)).$$

*Proof.* By Theorem 4.3 in Whitmeyer and Zhang (2022), to induce an attention strategy that is optimal for the principal, the relative incentives, such as $u(a, \omega) - u(a_1, \omega)$ for $a \in S, \omega \in \Omega$, are uniquely pinned down. Since the agent discounts the relative incentives by $\beta$, for the relative incentives to equal those of the principal, they need to be scaled up by $\frac{1}{\beta}$. This is achieved if and only if for all $\omega \in \Omega$, $u(a, \omega) - u(a_1, \omega) = \frac{1}{\beta}(v(a, \omega) - v(a_1, \omega))$. This in turn is equivalent to all elements $m_1, m_2$ of $Z$ that are directly connected by an edge (that is, are achieved at the same state) to satisfy $u(m_1) - u(m_2) = \frac{1}{\beta}(v(m_1) - v(m_2))$. $\square$

[21] This statement holds also under unknown bias.

[22] Under an unknown bias, as we have shown under complete feedback, regret and rejoicing are not only optimal because they minimize the variance of emotions but also because they allow tailoring the incentives to the state.

The above is an only-if statement. It ensures only that if we restricted the agent's choice to the consideration set $S$, a first-best attention strategy would be incentive-compatible (Whitmeyer and Zhang 2022). To implement the first-best attention strategy when the whole decision problem $A$ is feasible, it is sufficient to punish the actions in $A \setminus S$ with sufficiently negative emotions. As long as the principal deters the agent from choosing any action in $A \setminus S$, she is indifferent with regards to how strongly she punishes these actions as it does not affect her utility nor the implementation cost of realized emotions.

The utility difference being scaled up by $\frac{1}{\beta}$ is a transitive relation that extends to all nodes on the largest connected subgraph. If $z$ and $\tilde{z}$ are on the same connected subgraph of $Z$, then there is a finite chain of elements $(z_1, ..., z_k)$ of $Z$ such that $z = z_1$, $\tilde{z} = z_k$, and each adjacent pair is directly connected. If each of those directly connected pairs has a utility difference scaled up by $\frac{1}{\beta}$, then

$$
\begin{aligned}
u(\tilde{z}) - u(z) &= (u(z_k) - u(z_{k-1})) + ... + (u(z_2) - u(z_1)) \\
&= \frac{1}{\beta}(v(z_k) - v(z_{k-1}) + ... + \frac{1}{\beta}(v(z_2) - v(z_1)) = \frac{1}{\beta}(v(\tilde{z}) - v(z)).
\end{aligned}
$$

Conversely, if the utility difference of all pairs of nodes on the same connected subgraph is scaled up by $1/\beta$, then trivially the utility difference of directly connected nodes is scaled up by $1/\beta$.

Thus, an optimal attention strategy of the principal is implemented only if the utility difference of all elements on the same connected subgraph are scaled up by $1/\beta$. By Assumption 5, the sets of all messages that are consistent with some partition element $P \in \mathcal{P}$ are connected. By the definition of edges, such sets for different partition elements are not directly connected. In other words, $Z$ has one connected subgraph for each partition element $P \in \mathcal{P}$. This pins down the utility function $u$ except for one constant per subgraph, which is determined by the implementation-cost minimization problem. More specifically, by the above, we can write

$$
m(a, \omega) = \left(\frac{1}{\beta} - 1\right)\big(v(a, \omega) - v(\mathcal{P}(\omega))\big). \tag{27}
$$

The remaining problem is to determine the "reference point" $v(\mathcal{P}(\omega))$ for each partition cell.

The optimal reference point is connected to the concept of central tendency from statistics. The central tendency $\mathcal{C}_f(X)$ of random variable $X$ with respect to a strictly convex function $f$ is defined as

$$
\mathcal{C}_f(X) := \arg\min_c \mathbb{E}[f(X - c)],
$$

which is known to be unique under strict convexity of $f$. Here, we consider the central tendency $\mathcal{C}_\phi$ with respect to the strictly convex implementation cost $\phi$.

**Lemma 10.** *Under implementation cost $\phi(x) = |x|^k$ with $k > 1$, the implementation cost minimizing solution has reference point $v(P)$ equal to the central tendency of payoff $v$ on the partition element $P$ with respect to the optimal attention strategy.*

*Proof.* First, the centrality measure of emotions must be 0, otherwise we could decrease the implementation cost. Let $P \in \mathcal{P}$. The attention strategy that is implemented defines a distribution over emotions. The implementation cost on this partition is the expectation of $\phi(u(a,\omega) - v(a,\omega))$. Under the optimal emotions, $\phi(u(a,\omega) - v(a,\omega) - c)$ is minimized at $c = 0$, otherwise one could further reduce the implementation cost. This implies by definition that the central tendency of emotions is 0.

Next, we show two properties of the central tendency. First, $\mathcal{C}_\phi(X + b) = \mathcal{C}_\phi(X) + b$ due to

$$\arg\min_c \mathbb{E}[\phi(X + b - c)] = b + \arg\min_c \mathbb{E}[\phi(X - c)].$$

Second, under $\phi(x) = |x|^k$ with $k > 1$, $\mathcal{C}_\phi(aX) = a\mathcal{C}_\phi(X)$ for all $a \in \mathbb{R}$. First of all, note that $\phi(ax) = |ax|^k = (|a||x|)^k = |a|^k|x|^k = |a|^k\phi(x)$. Then,

$$\arg\min_c \mathbb{E}[\phi(aX - c)] = a \arg\min_c \mathbb{E}[\phi(a(X - c))] = a \arg\min_c \mathbb{E}[|a|^k\phi(X - c)] =$$
$$a \arg\min_c \left(|a|^k\mathbb{E}[\phi(X - c)]\right) = a \arg\min_c \mathbb{E}[\phi(X - c)].$$

Finally, using the two previous facts we show that the reference point of $P$ is the central tendency of the payoff $v$, with respect to the distribution induced by the implemented attention strategy on $P$. The reference point $v(\mathcal{P}(\omega))$ is the (hypothetical) payoff $\tilde{v}$ that obtains emotions 0 according to equation (27). By equation (27), emotions are a linear function of the payoff $v(a,\omega)$. Then, the realized payoff is also a linear function $f$ of emotions. The payoff $\tilde{v}$ is thus the image of 0 under $f$, $\tilde{v} = f(0)$. By the above, $\mathcal{C}_\phi(aX + b) = a\mathcal{C}_\phi(X) + b$, so $\mathcal{C}_\phi$ commutes with linear functions. By 0 being the central tendency of emotions, $\tilde{v}$ is the central tendency of the payoff. □

As is known, the central tendency $C_\phi(X)$ of random variable $X$ with respect to $\phi(x) = x^2$ equals the expectation of $X$. This concludes the proof.

# VII Appendix C: Interpretation as Contracting with an Expert

As mentioned in the main text, the model can be interpreted in a more conventional way as a principal hiring an agent (expert) to acquire information. Of particular relevance are CEO and fund manager compensation.

The principal reaps utility $v(a,\omega)$ from action $a$ under state $\omega$ and pays monetary transfers (e.g., wages) $m(a,\omega)$ to the agent, which are valued quasi-linearly by the agent. The bias of the expert can capture both a conflict of interest between principal and agent, as well as a behavioral bias of the expert. We discuss both cases in turn.

## A Expert with Conflicting Interests

Our model can be interpreted as the optimal monetary contract between a principal and an expert with a conflict of interest over how much information to acquire. In particular, assume that both principal and agent care about the information cost as well as the decision outcome but differ in how they weigh these considerations. For instance, investors or shareholders may internalize the time cost of information gathering to some degree due to the resulting decision delay or because it diverts the agent from other valuable activities. CEOs or fund managers may care about the decision outcome out of intrinsic motivation or reputational concerns. However, from the principal's perspective, the agent might overemphasize the information cost in the absence of financial incentives, potentially leading to suboptimal information acquisition. In these contexts, it is realistic to assume that information acquisition is unverifiable and only the outcome from decision-making is contractible. Information acquisition often does not produce tangible evidence and even when it does, the evidence might be hard to communicate or verify without the necessary expertise on the principal's side (Lindbeck and Weibull 2020). Finally, we take serious the possibility that the principal is uncertain about how much the agent overweights the information cost, by incorporating hidden information.[23]

The last model component to reinterpret is the cost of emotions $\phi$. Under a quadratic cost $\phi(x) = x^2$, our cost-of-emotions refinement is equivalent to an *ex-ante budget constraint* plus the refinement of reducing the variance of transfers. Note that the mean squared transfers equal the squared mean transfer plus the variance of transfers, as for any random variable $X$, $\mathbb{E}[X^2] = \mathbb{E}[X]^2 + \text{Var}[X]$. Thus, minimizing the quadratic cost is equivalent to minimizing the sum of the squared expected transfers and the variance of transfers. The principal can add a constant to transfers uniformly to reduce the expected transfers to 0, without affecting the variance of transfers. Further, this does not affect the agent's behavior. Thus, minimizing the quadratic cost $\phi$ is equivalent to an ex-ante budget constraint

$$\int \left( \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) m(a, \omega) \right) dF(\beta) = 0 \tag{BC}$$

and the refinement of minimizing the variance of transfers,

$$\int \left( \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) m(a, \omega)^2 \right) dF(\beta).$$

Under an interpersonal interpretation with $m$ as monetary transfers, $\Phi$ can be thought of as a implementation cost or as a cost of deviating from an ex-ante budget. Further, when interpreting the principal's problem in organizational contexts where monetary transfers are not used, we can interpret transfers as the reduced-form utility of reward- and punishment-actions taken by the

[23] A shortcoming of this application is that our model does not feature a limited liability constraint, which binds if the budget $b$ is low enough.

principal. Rewards, such as praise and gifts, as well as punishments, such as making the agent perform unfavorable tasks (which require monitoring), might both be costly for the principal, as captured by $\phi$.

## B Dynamically-Inconsistent Expert

If we interpret bias as a behavioral bias of the agent, then our optimal contract can be readily interpreted as the *welfare-maximizing* commitment contract for the agent, subject to an ex-ante *budget balance constraint*, if we again assume a quadratic cost $\phi(x) = x^2$.[24]

Furthermore, as demonstrated below, our model also applies to a *profit-maximizing* principal who hires a dynamically-inconsistent expert, subject to a *participation constraint.* Here, the principal values the decision outcome net of transfers paid to the agent, but does not care about the agent's attention cost directly. The agent, on the other hand, cares only about the transfers received net of the attention cost. Instead of a budget constraint, the principal faces an ex-ante participation constraint that guarantees the agent a certain expected utility before his bias realizes. The participation constraint effectively makes the principal internalize the agent's attention cost, rendering the problem equivalent to ours.

In these contexts, our work contribute to the understanding of compensation schemes, specifically those that assess the agent's performance relative to a peer benchmark. Relative performance evaluation is common in compensation schemes for fund managers (Ma, Tang, and Gómez 2019; Evans, Gomez, Ma, and Tang 2023) and, to some degree, for CEOs (Edmans, Gabaix, and Jenter 2017).[25] Such benchmarks can be justified by by Holmström's (1979) informativeness criterion in standard moral-hazard settings, because subtracting exogenous common shocks to the return of the industry increases the informativeness of the performance evaluation. Our model offers a different explanation based on incentives for information acquisition. Peer comparisons can be interpreted as counterfactual comparisons, since payoffs of relevant peers inform the principal about the payoff that alternative actions would have lead to. Indeed, Theorem 3 shows that under a known bias, transfers proportional to the difference between the realized return and the expected return under the realized market conditions are optimal.

Consider a principal who hires a dynamically-inconsistent agent (expert) to acquire information.[26] The principal reaps utility $v(a, \omega)$ from action $a$ under state $\omega$ and pays monetary transfers (wages) $m(a, \omega)$ to the agent. The agent maximizes transfers paid to them. Both have quasi-linear utility. After signing the contract, the agent faces a stochastic temptation shock to his preferences, as in random-indulgence models. Accordingly, the individual-rationality constraint needs to hold

[24] At first glance, a welfare-maximizing principal would have a different objective than our principal because she would care about the agent's transfers. However, this is immaterial as transfers cancel out in expectation due to the ex-ante budget constraint.

[25] Evans, Gomez, Ma, and Tang (2023) find that 71% of mutual funds in the US use peer-benchmarked compensation.

[26] See Kaur, Kremer, and Mullainathan (2015) for evidence on self-control issues at work.

in expectation. The principal's problem is

$$\max_{m,(p_\beta)_\beta} \int \left( -c(p_\beta) + \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega) p_\beta(a|\omega)\big(v(a,\omega) - m(a,\omega)\big) \right) dF(\beta)$$

s.t.

$$\forall\beta\colon\ \ p_\beta \in \arg\max_p -c(p) + \beta \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega) p(a|\omega) m(a,\omega) \tag{IC}$$

$$\int \left( -c(p_\beta) + \beta \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega) p_\beta(a|\omega) m(a,\omega) \right) dF(\beta) \geq U_0 \tag{IR}$$

If the agent was dynamically consistent ($\beta \equiv 1$), the principal would achieve the first-best by selling the firm to the agent. In the presence of a bias, the principal can be seen as maintaining control of the firm in order to provide a self-control mechanism to the agent for their mutual benefit.

By standard arguments, the (IR) constraint is binding. Inserting the (IR) constraint into the objective, we obtain the following equivalent formulation.

$$\max_{m,(p_\beta)_\beta} \int \left( -c(p_\beta) + \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega) p_\beta(a|\omega) v(a,\omega) \right) dF(\beta) - U_0$$

s.t.

$$\forall\beta\colon\ \ p_\beta \in \arg\max_p -c(p) + \beta \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega) p(a|\omega) m(a,\omega) \tag{IC}$$

$$\int \left( -c(p_\beta) + \beta \sum_{\omega\in\Omega}\sum_{a\in A} \mu(\omega) p_\beta(a|\omega) m(a,\omega) \right) dF(\beta) = U_0 \tag{IR'}$$

The constant $-U_0$ in the principal's objective can be ignored. Identifying $m$ with $v+m$ in the main text, the objective and (IC) constraint are identical. The (IR') constraint can be satisfied by adding a uniform constant to emotions, so (IR') acts in effect as an ex-ante budget constraint. Adding the refinement that the principal chooses the mechanism with lowest variance of transfers, renders the problem essentially equivalent to the principal's problem in the main text.[27] The only difference is that the optimal transfers emulate the payoff $v$ in addition to the emotions in the main formulation. That is because in this formulation the agent has no direct interest in the decision. Additionally, the optimal transfers are uniformly shifted by a constant, which depends on the outside option $U_0$.

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[27]The equivalence also holds under an appropriate symmetry constraint or when the principal chooses a menu of transfer schemes.

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